

Fitness drivers of division of labor in vertebrates

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eLife Assessment

This **useful** study develops an individual-based model to investigate the evolution of division of labor in vertebrates, comparing the contributions of group augmentation and kin selection. The model incorporates several biologically relevant features, including age-dependent task switching and separate manipulation of relatedness and group-size benefits. However, the evidence remains **inadequate** to support the authors' central claim that group augmentation is the primary driver of vertebrate division of labor. Key modelling assumptions—such as floater dominance advantages, the absence of task synergy, and the narrow parameter space explored—restrict the potential for kin selection to produce division of labor, thereby limiting the generality of the conclusions.

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Abstract

Although division of labor as a means to increase productivity is a common feature in animal social groups, most previous studies have focused almost exclusively on eusocial insects with extreme task partitioning. Empirical evidence of division of labor in vertebrates is scarce, largely because we lack a theoretical framework to explore the conditions under which division of labor is likely to evolve in cooperatively breeding systems where helpers remain capable of breeding throughout their lifetime. By explicitly considering alternative helping tasks with varying fitness costs, we model how individual decisions on task specialization may influence the emergence of division of labor under both direct and indirect fitness benefits. Surprisingly, we find that direct survival benefits of living in larger groups are the primary force driving the evolution of cooperation to enhance group productivity, and that indirect fitness benefits derived from related group members are only a non-essential facilitator of more stable forms of division of labor in cooperative breeders. In addition, we find that division of labor in vertebrates is favored by harsh environments. Ultimately, our model not only makes key predictions that are consistent with existing empirical data, but also proposes novel avenues for new empirical work in vertebrate and invertebrate systems alike.

Introduction

Division of labor as a means to boost productivity holds significance not only in human societies, but also those of other animals [1–5]. Nearly 250 years ago, the economist Adam Smith [2] proposed that division of labor via task specialization within human societies not only improves individual efficiency and group productivity, but also reduces the costs associated with switching tasks. Similarly, in nonhuman animal societies, division of labor to increase group reproductive output (analogous to economic gain) is characterized by group members specializing in particular tasks, either temporarily or permanently [6–8]. To date, most theoretical and empirical work on the selective pressures favoring division of labor via task specialization in animal societies has focused on “classically” eusocial insect societies characterized by permanently sterile workers that are never able reproduce [8–13]. Since workers in eusocial societies can, in most cases, only obtain fitness benefits indirectly through kin selection by maximizing their colony reproductive output, selection can occur at the colony level, typically resulting in very low conflict of interest between group members. However, complex societies occur outside of eusocial insects, such as in many birds and mammals, including humans. Because sterile workers are absent in vertebrates and other cooperative breeding societies, conflict can be much higher than in most eusocial insect societies. The few studies of division of labor outside of eusocial insects have primarily focused on reproductive division of labor (i.e., reproductive skew), despite that fact that temporal or permanent non-reproductive organisms can engage in an array of different helping tasks [3, 14, 15]. Given the scarcity of studies examining the selective advantages of division of labor beyond eusocial insects (but see [16]), our understanding of the evolutionary pressures of division of labor is largely confined to instances where task specialization is linked to sterility and where conflict of interest is generally low [17].

In cooperatively breeding vertebrates, which typically live in societies with more equitable sharing of reproduction than eusocial insects (i.e., low reproductive skew), low or mixed kinship, and often have unrelated helpers, conflict of interest can be high and direct fitness benefits may be more relevant in the evolution of division of labor than indirect benefits [18–20]. In fact, shared fitness incentives to increase group productivity are not limited to eusocial insects with high within-group relatedness and sterile workers. In cases where group size positively correlates with an increase in the survival or reproduction of group members, division of labor via task specialization may also evolve under direct fitness benefits to subordinate members of a group (i.e., group augmentation [21, 22]). Although a great deal of recent empirical work highlights the importance of direct benefits in the evolution of cooperative breeding behavior in vertebrates [23–26], we still lack an understanding of the joint influence of direct and indirect fitness benefits in the evolution of division of labor.

Understanding how direct and indirect benefits interact is particularly important in systems where individuals may differentially bear the fitness costs of cooperation. Most previous models of division of labor via task specialization considered all helping tasks under the same fitness umbrella, which is a good approximation in eusocial insect societies because colony members share fitness costs and benefits. In vertebrates and social insects with fertile workers, however, helpers may obtain fitness benefits directly via reproduction, and thus, selection occurs more strongly at the individual level than it does in most eusocial insects. Group members in vertebrate societies may then contribute to distinct helping tasks differently according to individual life history strategies [27–29] or differences in the magnitude or type of fitness costs and benefits that they receive from cooperative tasks [30]. One of the key distinctions between fitness costs is whether a task impacts more immediate survival or direct reproduction. For instance, defensive behaviors such as predator mobbing may have a high risk of injury and death [31–33], whereas other activities such as allofeeding or nest/territory maintenance may affect body condition owing to lower feeding rates and time and energy investment, consequently affecting

individual reproductive output [34–36]. Therefore, changes in life histories, such as those linked to the likelihood of becoming a breeder, may select for task specialization according to the relative direct fitness cost associated with different helping tasks.

In addition to life history differences, demographic, social, and ecological factors that affect help workload and the prospect of helpers becoming breeders are likely to have a strong impact on task allocation because they alter the probability of obtaining direct fitness benefits [37, 38]. For example, age is known to be an important factor influencing helping workload in vertebrates [39, 40], is often correlated with differences in other life-history traits [36, 41–43], and may be a proxy for other processes involved in the evolution of help [44, 45]. Moreover, age-dependent task allocation (i.e., temporal polyethism) is common in eusocial insects that do not show morphological differentiation [46]. Social and ecological factors are also thought to influence the emergence of division of labor in eusocial insects [47–49]. Indeed, high relatedness may reduce conflict of interest between group members and promote the evolution of division of labor [50–54]. Similarly, in harsh environments where group size strongly influences survival [55–58], division of labor may be favored to more efficiently increase group productivity.

To disentangle the role of direct versus indirect fitness benefits in the evolution of division of labor in cooperative breeding societies with totipotent helpers, we created an individual-based model in which we varied the influence of different evolutionary forces to understand under which social and ecological circumstances division of labor may arise to maximize the inclusive fitness of individuals in a group. We used distinct tasks with different fitness costs to maximize the group reproductive output when they were performed to a similar extent. Individuals in our model were allowed to evolve different behavioral responses or reaction norms in task specialization, depending on their probability of becoming breeders. We compared the results of this model to those of a benchmark model in which the reproductive outcome was independent of the type of task performed by group members. We present the results for different environmental conditions in the presence or absence of relatedness (indirect) or group size (direct) benefits. Ultimately, this work demonstrates that the direct fitness benefits of group living attained in increasingly harsh environments are key to the evolution of division of labor in vertebrates.

Results

Rationale

Our model consists of a population characterized by overlapping generations residing in a habitat featuring a fixed finite number of breeding territories. These territories are monopolized by groups composed of a single breeder and a variable number of subordinate helpers, the number of which is shaped by population dynamics, with all group members capable of reproducing during their lifetime. Individuals in the population fall into one of three categories: (1) *breeders* that monopolize reproduction; (2) *subordinates* that may perform different helping tasks, either in their natal group or in an outside group to which they dispersed; and (3) *floaters* that dispersed from their natal group but do not join another group.

We assume complete reproductive skew such that subordinate group members are prevented from reproducing at a given timestep and may only help raise the offspring of the dominant breeders. Dispersers cannot reproduce without acquiring a territory (denoted here as floaters). Helpers are not sterile and may inherit the breeding position when the breeder dies. When this occurs, helpers and dispersers from other groups compete for the breeding position and win with a probability proportional to their dominance value (Figure 1; Table 1). We use the term dominance value to designate the competitiveness of an individual compared to other candidates

in becoming a breeder, regardless of group membership. Dominance value increases as a function of age, serving as a proxy for resource holding potential (RHP), and decreases as a function of help provided, reflecting costs to body condition from performing working tasks (Eq. 2).

Subordinate helpers in our model may engage in one of two distinct tasks with differing fitness costs: (1) *defensive tasks* (i.e., defense of the dominant breeders' offspring at cost to their own survival); or (2) *work tasks* (i.e., tasks with an associated cost to their body condition and consequently to their competitiveness to become breeders in the future). Subordinate helpers can then evolve different workloads and exhibit task specialization that is either fixed throughout their lifetime or varies with their dominance value. The parameters x_h and y_h (Table 1) reflect how much the subordinates' survival and dominance values decrease with increasing investment in defense or work tasks, respectively. Although we do not assume an inherent difference in body condition between individuals when born, variance in age and helping propensities can create differences in the fitness costs linked to task performance.

We assume net survival benefits from living in large groups (i.e., group augmentation), which occurs in many social species [56, 57, 59, 60]. Subordinate members of a group may therefore gain both direct and indirect fitness benefits from helping to increase the productivity of breeders, if doing so results in an increase in group size. In the model, the benefit of alloparental care to the productivity of the breeder is greatest when group members provide overall similar workloads for both defensive and work tasks, since both are often needed for the survival of the offspring [61, 62]. For example, in many cooperatively breeding birds, the primary reasons that breeders fail to produce offspring are (1) starvation, which is mitigated by the feeding of offspring, here considered as a work task, and (2) nest depredation, which is countered by defensive behavior [61, 63]. Consequently, both types of tasks are often necessary for successful reproduction, and focusing solely on one while neglecting the other is likely to result in lower reproductive success than if both tasks are performed by helpers within the group. To compare between the helper's preferred task and the task needed to maximize group productivity, we created a benchmark model in which the breeder's productivity is not constrained by the type of help provided. Higher productivity translates to larger group sizes, which enhances the survival prospects of all group members. Hence, subordinates have an incentive to develop division of labor. Subordinates that remain in their natal group may also gain improved fitness by indirectly passing on their genes through kin. Both group size and relatedness in our model are emergent properties of a series of demographic processes, including mortality, dispersal, and help-dependent breeder fertility.

To distinguish between the effects of group augmentation and kin selection on inclusive fitness, we created two additional models. In the first model, we removed the benefits of group size (group augmentation) by setting the parameter $x_n = 0$ (Table 1). In the second model, we removed the effect of kin selection by simulating a cross-foster manipulation in which offspring just born are interchanged between different groups (see Methods for additional details). To assess how habitat quality affects task specialization and the emergence of division of labor, we systematically varied habitat quality across simulations. Harsh environments are those that have high mortality (i.e., $m = 0.3$; Table 1), which also raises the turnover of the breeding positions. Lower values of m reflect an increased probability of individuals surviving another breeding cycle and, therefore, the chances of habitat saturation. The steps in the breeding cycle of the model are illustrated in Figure 1.

Direct versus indirect fitness benefits

We first examined how both direct fitness benefits from group living and indirect fitness benefits from aiding kin influence the evolution of division of labor through task specialization across environments of varying quality. Our results suggest that voluntary division of labor involving tasks with different fitness costs is more likely to emerge initially because of direct fitness benefits (Figure 2; GA), but a combination of direct and indirect fitness benefits leads to higher rates and

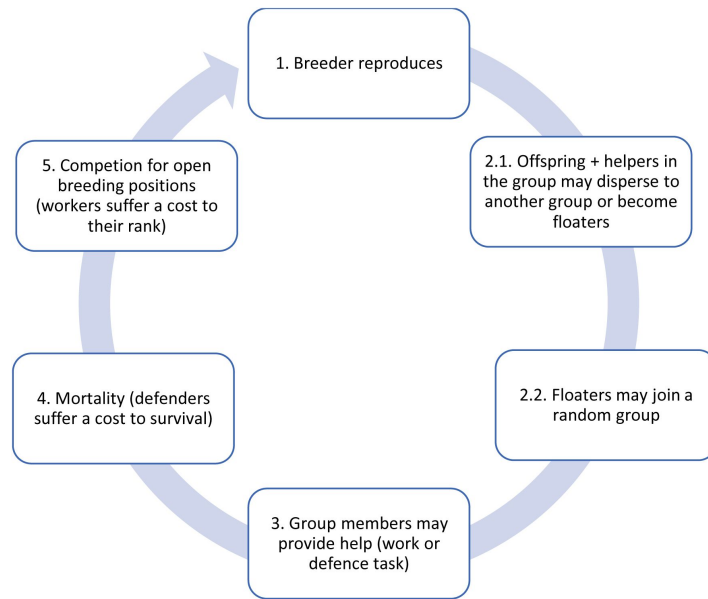


Figure 1.

Diagram of the scheduling executed per breeding cycle.

(1) A breeder reproduces. Its productivity depends on the cumulative level of brood care provided by the group during the previous breeding cycle. Maximum productivity is achieved when different helping tasks are performed to a similar extent. (2) Subordinates may disperse to become floaters, or they may stay in the group and help. Dispersers/floaters may join a random group to become subordinates. (3) Subordinates in the group (both natal and immigrant individuals) either work to provision to the breeder's offspring or display defensive forms of help. (4) Individuals survive contingent on group-living benefits and dispersal costs, as well as the cost of defensive activities. (5) If a breeder dies, helpers in the group and a sample of floaters compete for the breeding position. Individuals still alive ascend one age class, and the cycle starts all over (i.e., next breeding cycle).

Symbol	Meaning	Value
Genes		
α	Genetic propensity to help	0
β	Genetic predisposition to disperse versus remain in a group	1
β_0	Intercept in the dispersal reaction norm	1
β_R	Effect size of rank on dispersal propensity	0
γ_0	Intercept in the task specialization reaction norm	0
γ_R	Effect size of rank on task specialization	0
Scaling parameters		
y_h	Effect size of cost of help on dominance when performing “work tasks”	0.1
x_h	Effect size of cost of help in survival when performing “defensive tasks”	3 / 5 / 7
x_n	Effect size of the benefit of group size in survival	0 / 3
x_0	Intercept in the survival function	1.5*
m	Baseline mortality	0.1 - 0.3
f	Mean number of groups a floater samples for becoming a breeder	2*
k_m	Deviation from a perfect split in the need of both defense and work tasks	0.1
k_h	Effect size of the cumulative help of subordinates on the fecundity of the breeder	1
k_0	Fecundity of the breeder in the absence of help	1
μ	Mutation rate	0.05
σ_μ	Mutation step size	0.04
N_b	Number of breeding territories	5000
Phenotypic traits		
H	Level of help of either kind provided to the breeder $H = \alpha$, applying boundary of 0 for negative numbers. We express cumulative level of help given in a group as $\sum H_i$	
D	Dispersal propensity $D = \beta$, applying boundaries between 0 and 1	
R	Dominance value, applying boundary of 0 for negative numbers	
T	Probability of choosing “defensive task” vs “work task”	
S	Survival probability	
K	Fecundity of the breeder	
Variables		
t	Age	
N	Emergent group size	
N_f	Total number of floaters	
$N_{f,b}$	Number of floaters bidding for a breeding position, given by $N_{f,b} = f * N_f / N_b$	

Table 1.

Overview of notation.

Values conveyed for the genes are initial input values, values given for the scaling parameters are fixed throughout the simulations. If more than one value is given, results are shown to display the effect of this parameter’s variation. *: Additional values of these parameters are discussed in the Supplemental Material. A broader parameter landscape was explored but not included in the manuscript.

more stable forms of division of labor (**Figure 2**; KS + GA). It is important to note that, as depicted in **Figure 2**, intermediate values of task specialization indicate in all cases age/dominance-mediated task specialization ($\gamma_R \neq 0$; **Table 1**) and never a lack of specialization ($\gamma_R = 0$; **Table 1**); further details are shown in the next section: “*The role of dominance in task specialization*”.

In contrast to expectations from models of eusocial insects, our vertebrate model does not readily evolve division of labor when only kin selection is considered (see the section “*Kin selection and the evolution of division of labor*” and Figures S1-S3 in the Supplemental Material for more detail). Specifically, forms of help that impact survival never evolve under any environmental condition when only kin selection occurs (**Figure 2**; KS). This occurs because with kin competition for the breeding position and no survival benefits of group membership, the reduced incentives to remain in the natal group drive most subordinates to disperse and breed elsewhere (average dispersal rate between 88% in harsh environments and 93% in benign environments; **Table S1**). Adding a reaction norm of dispersal to dominance rank did not change the results (see the section “*Dominance-dependent dispersal propensities*” and **Figure S4** in the Supplemental Material). The few subordinates that remain in the natal territory help to obtain indirect benefits (average within-group relatedness between 0.36 in benign environments and 0.72 in harsh environments; **Table S1**). Even though individuals in our model cannot adjust their help according to their relatedness to the breeder, the high levels of relatedness that evolved under this paradigm indicate that allowing for targeted help would not qualitatively change the results. Survival costs of defensive tasks are then avoided, as mortality without group benefits is relatively high, while costs to dominance value are small for close relatives because of indirect fitness benefits and low conflict of interest (**Figure 2**; KS). Adjusting input parameters to obtain higher survival probabilities and favoring philopatry did not qualitatively change the results, except for extreme values of baseline survival (see Supplementary Material), which indicates that kin selection alone is unlikely to select for the evolution of defensive tasks and division of labor in vertebrates.

Conversely, under only direct fitness benefits (group augmentation), tasks associated with costs to the dominance value are costlier because the only path to obtain fitness benefits is via direct reproduction. Therefore, defensive tasks that do not affect the dominance value are favored (**Figure 2**; GA). Under this scenario, division of labor can evolve to maximize the reproductive output of the breeders (**Figure 2**; GA, $m = 0.3$, $x_h \leq 5$), but also because it is the optimal strategy (**Figure 2**; GA, $m = 0.3$, $x_h = 7$). In benign environments, however, no help evolves due to the lower incentives to increase survival and the steep competition for reproduction resulting from habitat saturation.

When subordinates can obtain both direct and indirect fitness, division of labor is more strongly selected in increasingly harsher environments to increase breeder productivity (**Figure 2**; KS + GA). Overall, in moderate and harsh environments, work tasks with costs to the dominance value are more strongly selected than in benign environments due to high mortality increasing both the breeder’s turnover, reducing breeding competition (lower effective costs of work tasks), and the costs to survival (high costs of defensive tasks). However, in more benign (and often highly productive) environments that lead to habitat saturation, help likely evolved initially in family groups, and defensive tasks are favored because competition for the breeding position is lower under kin selection. In this scenario, kin selection acts not only by favoring subordinates in their natal group to increase the productivity of a related breeder (**Figure 2**; KS), but also by the increased survival of their siblings when augmenting the group size (**Figure 2**; KS + GA), regardless of whether they are related to the current breeder.

Overall, division of labor seems to be more likely to evolve in harsh environments where raising breeder productivity effectively enhances the survival of both helpers and relatives (**Figure 2**). In contrast, help evolves in benign environments largely due to kin selection benefits and subordinates performing their preferred task of defending the breeder’s offspring (**Figure 2**).

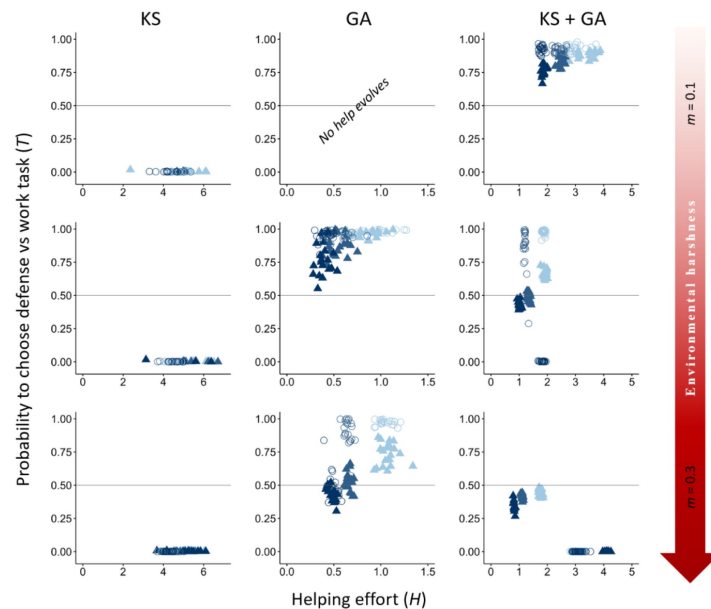


Figure 2.

Effect of environmental quality on alloparental care and division of labor.

The evolutionary equilibria for phenotypic levels of helping and task specialization are shown at three different levels of environmental quality, ranging from benign ($m=0.1$) to harsh ($m=0.3$), and for three different levels of cost of help on survival (light blue, $x_h=3$; blue, $x_h=5$; and dark blue, $x_h=7$), across 20 replicas. The vertical axis expresses the probability of individuals choosing a defensive task with a cost to survival versus a work task with a cost to their dominance rank. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$) or when no restrictions were introduced to the task performed by the group members ($\circ$). In each environment, additional details are given on the selective forces that play a role in the evolution of help and task specialization: help can only evolve by kin selection (KS), group augmentation (GA), or both (KS + GA). Additional details are provided in [Table S1](#). All input parameter values are described in [Table 1](#).

The role of dominance in task specialization

In the previous section, we explored the role of direct and indirect fitness benefits in the evolution of division of labor via task specialization. However, determining whether parameter γ_R (i.e., the direction and rate of change in task specialization with dominance) significantly differs from zero is crucial. Intermediate values of phenotypic task specialization may indicate division of labor among different age- or dominance-related groups (if $\gamma_R \neq 0$), or lack of specialization in either task (if $\gamma_R = 0$). Hence, for age-polyethism or temporal division of labor to arise in our model, individuals must switch between tasks based on their age-related dominance (probability of choosing a “defensive task” versus a “work task” $T \neq 0$ or 1, and $\gamma_R \neq 0$; [Table 1](#)). Across all scenarios, individuals are selected to perform tasks based on their probability of inheriting the territory ([Figure 3A](#)). Specifically, when subordinates prefer defensive tasks ([Figure 3A](#), $m < 0.2$, circles), they engage in work tasks at a young age and increasingly undertake defensive tasks as their dominance rises ([Figure 3](#), $m < 0.2$). Similarly, if their preference leans toward work tasks ([Figure 3A](#), $m \geq 0.2$, circles), younger subordinate helpers and those of lower dominance focus more on defense initially, but switch to work tasks as they approach the breeding stage ([Figure 3](#), $m \geq 0.2$). In essence, subordinate helpers allocate more time to tasks with lower fitness costs as they near a competitive dominance value, all without any active enforcement by other group members, resulting in age- or rank-polyethism. These results are consistent with the rise in division of labor, defined as within-individual consistency and between-individual differences in task choice [[64](#)].

Discussion

Our findings suggest that the direct benefits of group living play a driving role in the evolution of division of labor via task specialization in species with totipotent workers, including all vertebrates and a few insects. In contrast, in eusocial species characterized by high relatedness and permanent worker sterility, such as in most eusocial insects, workers acquire fitness benefits only indirectly, and so, their fitness is contingent upon the overall success of the colony rather than the individual [[65](#), [66](#)]. Group members in such eusocial species are therefore predicted to maximize colony fitness due to the associated lower within-group conflict, and to favor the evolution of division of labor to increase group productivity efficiency [[51](#), [52](#), [54](#)]. However, in taxa with lower within-group relatedness, such as nearly all vertebrates, division of labor is facilitated when cooperation is crucial for survival and when the benefits of cooperation are evenly distributed among group members [[3](#)]. In particular, survival fitness benefits derived from living in larger groups seem to be key for the evolution of cooperative behavior in vertebrates [[25](#), [67](#)], and may also translate into low within-group conflict. This suggests that selection for division of labor in vertebrates is stronger in smaller groups, where increased productivity significantly enhances survival, as opposed to research on eusocial insects that posits division of labor is more likely to evolve in larger groups [[68](#)–[70](#)]. That is not to say, however, that kin selection is unimportant in cooperatively breeding vertebrates or insects. Our model demonstrates that cooperative breeders are more likely to evolve division of labor when direct and indirect fitness benefits act in concert. Moreover, our model predicts that the type of task that individuals prefer depends on the type of fitness benefit that subordinates can attain. While direct fitness benefits in the form of group augmentation select more strongly for defensive tasks, kin selection alone seems to select only for work tasks. In groups of unrelated individuals where members can only gain direct fitness benefits, costs to dominance value and to the probability of attaining a breeding position are comparatively larger than survival costs. Conversely, groups with no group benefits tend to be small and suffer high mortality, for which defensive tasks only aggravate the situation.

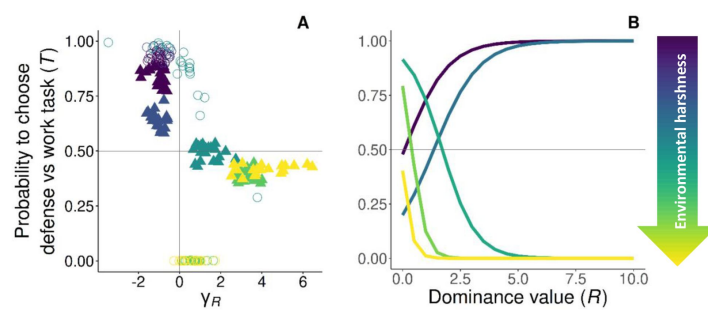


Figure 3.

Evolved reaction norms to age on the display of task specialization.

The evolutionary equilibria for the reaction norms of task specialization are shown at five different levels of environmental quality, ranging from benign ($m=0.1$, purple) to harsh ($m=0.3$, yellow), and $x_h=5$. **A:** $\gamma_R > 0$ signifies that individuals increase the probability of performing work tasks with dominance (defense $\rightarrow$ work), whereas $\gamma_R < 0$ signifies that individuals increase the probability of performing defensive tasks with dominance (work $\rightarrow$ defense). The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$), or when no restrictions were introduced to the task performed by the group members ($\circ$). Results show that at equilibrium when division of labor evolves, individuals increase the probability of performing their preferred task (circles), when increasing their dominance value. **B:** Evolved reaction norms to dominance value with average γ_0 and γ_R across 20 replicas for varying quality environments. All parameter values described in [Table 1](#).

Previous models of task allocation in eusocial insects have focused on how seemingly identical individuals of the same age and broad morphology may specialize in different types of tasks [8, 64]. In contrast, our model shows that extrinsic ecological factors are also important for the evolution of task specialization in addition to intrinsic factors such as age or dominance rank. In short, our model predicts that task specialization, which varies with age or rank, will depend on environmental quality. In more favorable environments, helpers are predicted to increase defensive tasks with age or rank, whereas in harsh environments, work tasks are predicted to increase with age or rank. Furthermore, harsher environments exert stronger selective pressure for task specialization due to increased survival challenges. Enhanced cooperation and improved group productivity through helping mitigate the negative effects of the environment. This finding is consistent with previous research on the role of ecological constraints in driving the evolution of cooperation [71–73].

Although rare, evidence of division of labor in cooperative breeding vertebrates comes from a few studies in mammals [74], birds [75, 76], and fishes [77, 78]. Our model predictions are broadly consistent with empirical data from many of these studies. For example, support for the role of environmental harshness on the evolution of division of labor via age-dependent task specialization in vertebrates comes from arguably the most similar vertebrate system to eusocial insects, the naked mole-rat (*Heterocephalus glaber*). Naked mole-rats live in self-dug burrows that protect their inhabitants from predation and climatic extremes [74]. Therefore, even though they inhabit habitats with scarce and unpredictable rainfall, environmental harshness is likely to be limited in their stable subterranean burrows, with mortality mainly spiking during dispersal attempts. Model predictions for species that evolved in benign environments are then met, with individuals shifting from work tasks such as foraging for food, digging, and maintaining the burrow system, to defensive tasks such as guarding and patrolling as individuals grow older and larger [74, 79, 80]. Although naked mole-rats exhibit high levels of within-colony relatedness [81], group size also exerts a positive influence on survival, potentially generating group augmentation benefits [82, 83]. Under these conditions, our model predicts the highest levels of task partitioning and division of labor. In contrast, *Neolamprologus pulcher*, a cooperatively breeding cichlid fish that lives in harsh environments with high risk of predation [62], shows the opposite tendency. In line with our model predictions, larger and older helpers of this species invest relatively more in territory maintenance, whereas younger/smaller helpers defend the breeding shelter of the dominant pair to a greater extent against experimentally exposed egg predators [78]. Territory maintenance has been shown to greatly affect routine metabolic rates and, hence, growth rates [84], which directly translates into a decrease in the probability of becoming dominant and attaining breeding status. In addition, both group augmentation and kin selection have been found to play a role in the evolution of help in this species [23, 55]. More research is needed to verify whether this trend is widespread in cooperatively breeding vertebrates with division of labor. The preferred form of help provided by individuals throughout their life course might deviate from our expectations if helpers exhibit differing abilities to perform distinct tasks at different ages or life stages. We assumed that individuals of all ages possessed equal capability to perform different tasks. However, as individuals age and their size and body mass increase, they may become more adept at certain tasks such as deterring predators [36, 85–89]. Body condition that is unrelated to age may also influence helper contributions, as individuals in poorer condition may be more constrained on their energy allocation in activities other than self-maintenance [90–92]. Furthermore, our model does not capture instances in which help in general or specific tasks are enforced by the dominant pair or other more dominant group members [93–95].

Despite evidence of task specialization and division of labor across several vertebrate taxa, other empirical studies have failed to find similar results. For instance, researchers failed to find evidence of task specialization in Damaraland mole-rats (*Fukomys damarensis*), despite their social structure resembling that of naked mole-rats [96, 97], with some variation in digging behavior with age and sex [98]. However, it is worth noting that researchers have only

investigated correlates between different work tasks, potentially overlooking negative correlations with other helping activities that pose a more direct cost to survival. In fact, many studies that reported an absence of division of labor did not consider defensive tasks in their analyses [96, 97, 99, 100]. Nonetheless, studies exist in which researchers did not find a negative correlation between defense and offspring provisioning or territory maintenance [42, 101, 102]. However, individuals may exhibit a positive or non-relationship between tasks if they also vary in total helpfulness across their lifetime, but may still switch the proportion of time/effort allocated to different types of tasks. In our model, we addressed for this potential tendency by restricting individuals from changing their overall helpfulness with age/dominance rank. We encourage future researchers to consider the proportion of time/energy investment in addition to examining correlations between tasks.

In summary, our study helps to elucidate the potential mechanisms underlying division of labor between temporal non-reproductives via task specialization in taxa beyond eusocial organisms. Harsh environments, where individuals can obtain direct fitness benefits from group living, favor division of labor thereby enhancing group productivity and, consequently, group size. Variation in the relative fitness costs of different helping tasks with age favors temporal polyethism, with individuals changing their behavior patterns to improve their chances of becoming breeders. Hence, future empirical research on division of labor should prioritize cooperatively breeding species inhabiting relatively harsh environments where subordinates benefit greatly from group membership, such as protection or improved food acquisition. Additionally, we emphasize the importance of including sentinel and defense behaviors in future studies to encompass a broader spectrum of fitness costs of cooperation, and to investigate potential changes in task investment throughout the helpers' lifetimes.

Materials and methods

We developed an individual-based model in which individuals may display and vary their efforts in different helping tasks. The ancestral state features an absence of alloparental care and no task specialization. We assume that residing in larger groups yields overall survival advantages, potentially resulting in direct fitness benefits associated with the increase in group size, as proposed by the group augmentation hypothesis. We also include the coevolution of the helping strategies with their dispersal propensity, as dispersal affects group size and kin structure, and thus may impact the strength of indirect fitness benefits.

Breeding cycle

We model a population consisting of a fixed number of breeding territories, each comprising a dominant breeder that reproduces asexually and monopolizes reproduction, as well as an undefined number of subordinates that queue for the breeding position. The number of subordinates is determined by the productivity of the breeders, the dispersal decisions of the group members, and immigration. Subordinates may help increase the fecundity of the breeder, so that the breeder's fecundity depends on the cumulative level of help provided by the subordinates within the group, with an optimum increase when different helping tasks are performed to a similar degree.

Offspring inherit the genes for dispersal and helping tendencies from the breeders, potentially exhibiting slight variation in the presence of mutations (**Figure 1**, Step 1). Each genetic trait is controlled by a single locus, and may take any real number. Mutations occur at each genetic locus at a frequency of μ , causing minor adjustments to the allele's value inherited from the parent. This alteration involves the addition of a value drawn from a normal distribution characterized by a mean of 0 and a standard deviation of σ_μ (**Table 1**).

After breeders reproduce, their offspring and the subordinates in the group may disperse independently from each other. Dispersal propensity is controlled by a gene with values ranging from 0 to 1, where 0 represents certain philopatry and 1 represents absolute dispersal ([Table 1](#)). Dispersers may migrate to another group to become subordinates or remain as floaters waiting for breeding opportunities, which is also controlled by the same genetic dispersal predisposition as subordinates ([Figure 1](#), Step 2). Successful dispersers enter the breeding queue within the group based on their dominance value rather than starting at the bottom of the hierarchy. Subordinate group members, either natal individuals born in the group or immigrants to the group, express some level of help in the form of alloparental care, which can potentially evolve to zero ([Figure 1](#), Step 3). To be conservative, we use the term help as synonymous with altruistic behavior, which increases the fitness of the beneficiary (i.e., breeder) with an immediate decrease in the actor's fitness. Hence, cooperative behaviors such as defense from adult predators that incur immediate benefits to the actor as well as by-product mutualistic benefits to other group members, are not considered in this study. Dispersal propensity, alloparental care provisions, and the type of helping task they perform all depend on their genetic predisposition. They may perform tasks that reduce their body condition and, therefore, their chances of becoming breeders, such as feeding offspring or territory maintenance (hereafter called “work tasks”), or they may perform tasks that reduce their immediate survival such as predator defense (hereafter called “defensive tasks”) [[35](#)]. For simplification, we assume complete dichotomous, short-term non-cumulative fitness consequences between the different types of tasks (i.e., survival and dominance costs are only sustained during the breeding season in which they perform the task, and individuals only perform one of the tasks in the given cycle). We note that the names given for each type of task are meant to facilitate interpretation, and that in some species, some tasks classified as work tasks may have a higher immediate cost in survival than in the probability to breed, and vice versa. However, helping tasks should be classified in empirical systems according to the primary fitness cost associated with each type of helping activity (e.g., if a work activity mostly negatively impacts survival in a given species, this task is described in this paper as defensive).

Individual survival depends on environmental quality, group membership, group size, and the amount of help provided during defensive tasks to the breeder's offspring ([Figure 1](#), Step 4). Environmental quality is expressed as a varying maximum survival probability, independent of social factors. Additional environmental effects, such as variation in mortality linked to dispersal, or on the probability of floaters finding a new group to start breeding, have been explored elsewhere [[67](#)].

If the breeder in a group dies, all subordinates in the group as well as a randomly drawn sample of floaters from outside the group compete for the breeding position ([Figure 1](#), Step 5). The number of floaters bidding for an empty breeding spot is then proportional to the relative abundance of floaters with regard to the total number of breeding territories, to account for spatial viscosity and the probability of being accepted in a new group ([Table 1](#)). In this model, age influences the probability of attaining the breeding position. Older individuals, owing to their higher resource holding potential, have a greater probability of becoming breeders, as observed in various species [[62](#), [103](#)–[111](#)]. Age in our model corresponds to the number of breeding cycles or seasons that an individual survives. Additionally, work tasks reduce an individual's dominance value as a result of the impact on their body condition. The probability of filling an empty breeding position is then implemented as a lottery weighted by the dominance value of the candidates. Therefore, subordinates may stay in the natal group and queue to inherit the dominant position [[105](#), [112](#)–[115](#)], but they may also disperse and breed or queue in another group [[37](#), [113](#), [116](#)]. We ran simulations for 200'000 breeding cycles or until equilibrium was reached for all genetic traits, across 20 replicas to assess consistency.

Age-dependent task specialization

The probability of choosing defensive versus work tasks T takes a logistic function with boundaries between 0 and 1 as given in Eq. 1 [\[67\]](#), in which individuals may adjust their helping task as they age and become more dominant,

$$T = \frac{1}{1 + \exp(-\gamma_R R - \gamma_0)} \quad (1),$$

where γ_R modifies the strength and direction of the effect of dominance (R) on task choice, and γ_0 acts as the intercept. Therefore, when T approaches zero, individuals specialize in work tasks throughout their lives, and when T approaches one, they specialize in defensive tasks. If $\gamma_R \neq 0$, given that $T \neq 0$ and $T \neq 1$, individuals show age-dependent task specialization.

In addition, the phenotypic expression of help (H) is regulated by another gene and equals the allelic value of gene α , which remains fixed throughout the helper's life. Even though in nature individuals are likely to change the total amount of help given throughout their lives, we make this simplification to allow easier interpretation of how individuals may switch between task types as they age. How individuals in the model may adapt their level of help with age and social and environmental conditions has been described elsewhere [\[67\]](#). Hence, the gene α regulates the amount of help expressed, while the genes γ determine which specific helping tasks are performed at different time points in the breeding cycle. The fitness costs of helping are described in the following section.

Fitness costs of different helping tasks

Helping tasks are classified in our model according to their short-term fitness consequences. We assume that work tasks incur a more immediate cost to the body condition of helpers because of expenses in terms of time and energy. Work tasks reduce the dominance value R to inherit the breeding position in our model following Eq. 2 [\[67\]](#), where γ_h is a parameter that influences the strength of the cost of help (H_{work}) in the dominance value. We assume that dominance is influenced by age (t) as a proxy for resource holding potential, with older individuals having a greater dominance value,

$$R = t - \gamma_h H_{work} \quad (2).$$

Note that the cost to an individual's dominance value is influenced by their current age and the current specific task that they are performing, the cost of which is not cumulative over time. Therefore, an individual's effective rank within the dominance hierarchy is, irrespective of whether they belong to a group or not, determined by their dominance value relative to other competitors for the breeding position, with higher dominance values corresponding to higher ranks in the hierarchy.

In contrast, we assume that defensive tasks have a more immediate fitness cost on survival (e.g., predator or space competitor defense) due to the high risk of injury. Therefore, subordinate helpers that defend have a survival probability (S_H) that decreases with the amount of help provided ($H_{defense}$) given by the logistic Eq. 3 [\[67\]](#), where x_h is a parameter that influences the strength of the cost of help in survival,

$$S_H = \frac{1-m}{1 + \exp(-x_0 + x_h H_{defense} - x_n N)} \quad (3.1).$$

In addition to defensive tasks, social and ecological factors also influence the survival of group members. The term m denotes the baseline mortality that determines how harsh the environment is. Higher values of m indicate higher overall baseline mortality for all individuals in the

population, irrespective of social factors. For simplicity, we assume that the harshness of the environment only impacts survival, although other effects such as on reproductive output or the strength in which help improves fecundity are likely to have an effect in nature [117]. Group size (N) provides survival benefits, such as those offered by safety in numbers or increased resource defense potential, for both subordinates and breeders (Eq. 3.1 and 3.2 respectively), where x_n is a parameter that quantifies the effect size of the benefit of group size in survival, and x_0 an intercept. Floaters sustain the highest mortality (S_F , Eq. 3.3) since they do not benefit from group protection,

$$S_B = \frac{1-m}{1+\exp(-x_0-x_n N)} \quad (3.2) \text{ and}$$

$$S_F = \frac{1-m}{1+\exp(-x_0)} \quad (3.3).$$

Need for division of labor

To assess the rules governing task specialization and division of labor, we first outline a basic model where individuals evolve their preferred helping task. Then we compare this to another model in which the breeder's reproductive outcome is maximized when the group's helping effort in each kind of tasks is performed to a roughly equal degree. Thus, in the second scenario, the maximum amount of cumulative group help of each type (H_{max}) that can affect fecundity is given by Eq. 4:

$$H_{max} = \frac{\sum_{i=1}^n H_{defense_i} + \sum_{i=1}^n H_{work_i}}{2} + k_m \quad (4),$$

where k_m is a parameter that relaxes the need for division of labor. The breeders then reproduce asexually. The quantity of offspring produced (K) is contingent upon the breeders' baseline fecundity (k_0) and the collective amount of help contributed by the helpers in the group. As indicated in the Eq. 5, the increase in fecundity shows diminishing returns as help increases, where k_h is a scaling parameter that modifies the benefit of the cumulative help provided by all the subordinates to increase the fecundity of the breeder,

$$K = k_0 + \frac{k_h(\sum_{i=1}^n H_{defense_i} + \sum_{i=1}^n H_{work_i})}{1 + \sum_{i=1}^n H_{defense_i} + \sum_{i=1}^n H_{work_i}} \quad (5).$$

If the cumulative defense effort is larger than H_{max} ($\sum_{i=1}^n H_{defense_i} > H_{max}$) or the cumulative work effort is larger than H_{max} ($\sum_{i=1}^n H_{work_i} > H_{max}$), then we use the value H_{max} instead. The breeder's fecundity is determined by a randomly selected value drawn from a Poisson distribution with mean of K .

Fitness benefits of helping

We include both direct and indirect fitness benefits of helping to assess the relative importance of each in the evolution of task specialization. Indirect fitness benefits from relatedness among group members (i.e., kin selection) emerge from demographic factors such as dispersal tendencies, mortality rates, and group dynamics, though they were not coded explicitly. To calculate the coefficient of relatedness between the breeder and subordinates in a group, we calculated the coefficient of a linear regression between the allelic values of the breeders and helpers of a neutral gene that changes exclusively by genetic drift [118, 119]. Direct benefits may also be obtained by enhancing group size by helping to increase the breeder's fecundity, as larger groups benefit from higher survival probabilities in our model (i.e., group augmentation).

To distinguish the effect of kin selection from group augmentation, we created two parallel models for comparison to the default model (group augmentation and kin selection acting independently), in which one of the fitness benefits was removed. In the "only group augmentation"

implementation, individuals just born (i.e., age = 1) that decide to stay in the natal territory as subordinates (**Figure 1** [↗](#), Step 2) are shuffled into another group, thereby removing the influence of relatedness from the model. In the “only kin selection” implementation, we removed the survival benefits linked to living in a group ($x_n = 0$).

Data and materials availability

Custom code used to generate the simulations is available at https://github.com/IreneGR92/Task_specialization [↗](#)

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Additional information

Author Contributions

conceptualization IGR & DR, methodology IGR, investigation IGR, visualization IGR, supervision DR, writing—original draft IGR, writing—review & editing DR.

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- Irene García-Ruiz

Supporting information

Kin selection and the evolution of division of labor

To assess the robustness of our findings on the evolution of division of labor driven by kin selection independent of group survival benefits, we further examined the influence of survival, dispersal, relatedness, and breeding opportunities outside the natal group.

One factor contributing to defense not readily evolving under kin selection alone is that the cost to dominance is relatively small when competing against relatives. Another possible explanation is that without group benefits, mortality rates are too high for survival-costly helping tasks such as defense to evolve. In **Figure S1** [↗](#), we examine the effects of increasing the baseline survival from $x_0 = 1.5$ (as in **Figure 2** [↗](#)) to $x_0 = 4.5$ (which simulates the addition of a group benefit of $x_n = 3$ to the baseline survival) and $x_0 = 10$ (an extreme scenario that greatly reduces the survival cost of defense tasks; see *Eq. 3* [↗](#).1).

Our results indicate that even with high baseline survival, kin selection strongly favors work over defense tasks. However, in very harsh environments ($m = 0.3$), where the survival costs of defense tasks are minimal ($x_0 = 10$ and $x_n = 3$; **Table S2** [↗](#)), some degree of task specialization and division of labor emerges (**Figure S1** [↗](#)). Despite this, most individuals still choose to disperse and breed

outside their natal territory when no group survival benefits exist (**Table S2** [↗](#)). This suggests that, without such benefits, offspring have little incentive to stay and help due to reduced reproductive opportunities and competition with kin. To test this, we eliminated the dispersal advantage by reducing the number of sampled groups from two ($f = 2$) to one ($f = 1$), incentivizing individuals to queue for breeding in their natal territory (**Figure S2** [↗](#)).

Even when increased philopatry, resulting from reduced outbreeding opportunities for floaters, potentially enhances the indirect fitness benefits of helping and task partitioning, division of labor only evolves when survival costs are exceedingly low ($x_0 = 10$; **Figure S2** [↗](#); **Table S3** [↗](#)). This suggests that while kin selection can promote the evolution of division of labor, it does so only under highly restrictive conditions with minimal survival costs. Therefore, in natural settings, other selective forces, such as direct fitness benefits, are more likely to drive the evolution of division of labor.

The strength of kin selection is likely more pronounced in our model than in natural populations due to the assumption of asexual reproduction. To assess whether the results remain valid with lower within-group relatedness, we shuffled half of the newly born individuals (i.e., age = 1) who choose to stay in their natal territory as subordinates, similar to the “only group augmentation” implementation. The results indicate that a reduction in relatedness increases the probability of individuals favoring defense over work tasks, but the overall conclusions of our model remain unchanged (**Figure S3** [↗](#)).

Dominance-dependent dispersal propensities

To assess the role of dominance or RHP on dispersal propensities D , and its potential effect on task specialization, we added a reaction norm to dominance value (R),

$$D = \frac{1}{1 + \exp(-\beta_R R - \beta_0)} \quad (\text{S1}),$$

where β_R modifies the strength and direction of the effect of dominance on dispersal, and β_0 acts as the intercept. This change allows individuals to adjust their dispersal and immigration propensities to their competitiveness, as well as their preferred helping tasks.

We found that incorporating a dispersal reaction norm to rank did not qualitatively influence the results on the evolution of division of labor and task specialization (**Figure S4** [↗](#)), with the conclusions remaining broadly unaltered.

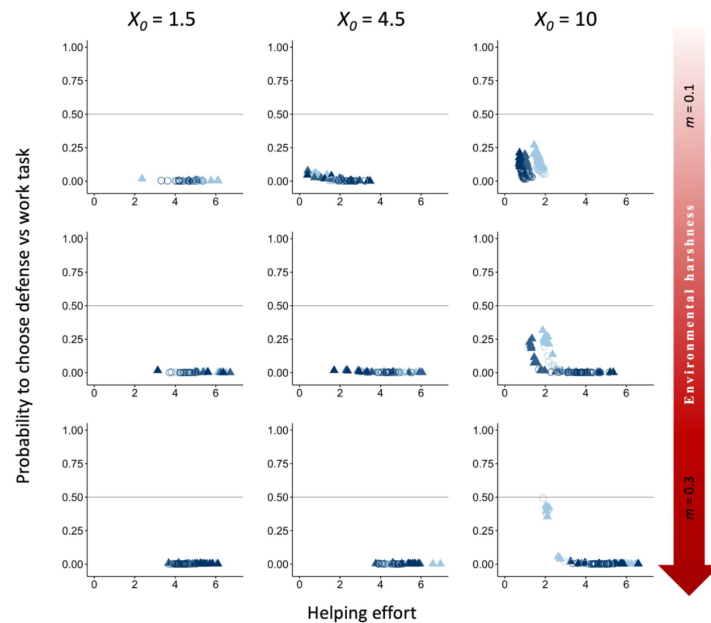


Figure S1.

Effect of increasing the baseline survival x_0 to favor the evolution of division of labor under only kin selection.

The evolutionary equilibria for levels of helping and task specialization are shown at three different levels of environmental quality, ranging from benign ($m = 0.1$) to harsh ($m = 0.3$), and for three different levels of cost of help on survival (light blue, $x_h = 3_{\text{environmental}} = 5$; and, ranging $x_h = 7$). The vertical axis expresses the probability of individuals choosing a defensive task with a cost to survival versus a work task with a cost to dominance. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$), or when no restrictions were introduced to the task performed by the group members ($\circ$). Input parameters are the same as in [Figure 2](#) (where $x_0 = 1.5$) except for $x_0 = 4.5$ and $x_0 = 10$ (higher survival for all individuals irrespective of group membership or environment; [Table 1](#)). Additional details are provided in [Table S2](#).

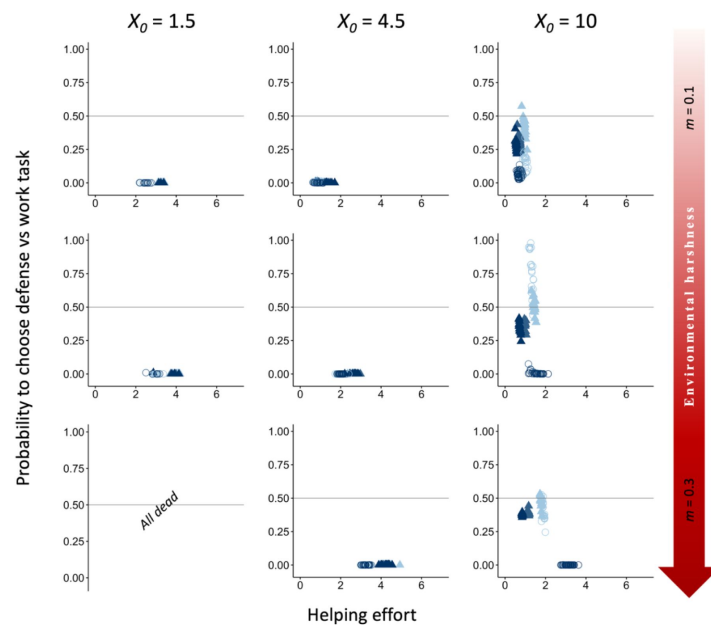


Figure S2.

Effect of reducing the incentives to disperse to favor the evolution of division of labor under only kin selection.

An increase in incentives to remain philopatric was achieved by reducing f to 1 ($f = 2$ in [Figure 2](#); [Table 1](#)). The evolutionary equilibria for levels of helping and task specialization are shown at three different levels of environment quality that range from benign ($m=0.1$) to harsh ($m = 0.3$), and for three different levels of cost of help on survival (light blue, $x_h = 3$; blue, $x_h = 5$; and dark blue, $x_h = 7$). The vertical axis expresses the probability of the individuals to choose a defensive task with cost to survival versus a work task with a cost to dominance. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (▲), or when no restrictions were introduced to the task performed by the group members (○). Other input parameters are the same as in [Figure 2](#) (where $x_0 = 1.5$) except for $x_0 = 4.5$ and $x_0 = 10$ (higher survival for all individuals irrespective of group membership or environment; [Table 1](#)). Additional details are provided in [Table S3](#).

Figure S3.

Effect of reducing within-group relatedness by half to mimic sexual reproduction.

A reduction in within-group relatedness was achieved by shuffling half of the philopatric newborns to another group in the KS and KS+GA implementations, and all for the GA implementation. The evolutionary equilibria for levels of helping and task specialization are shown at three different levels of environment quality that range from benign ($m=0.1$) to harsh ($m=0.3$), and for three different levels of cost of help on survival (light blue, $x_h = 3$; blue, $x_h = 5$; and dark blue, $x_h = 7$). The vertical axis expresses the probability of the individuals to choose a defensive task with cost to survival versus a work task with a cost to dominance. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (▲), or when no restrictions were introduced to the task performed by the group members (○). Input parameters are the same as in Figure 2. Additional details are provided in Table S4.

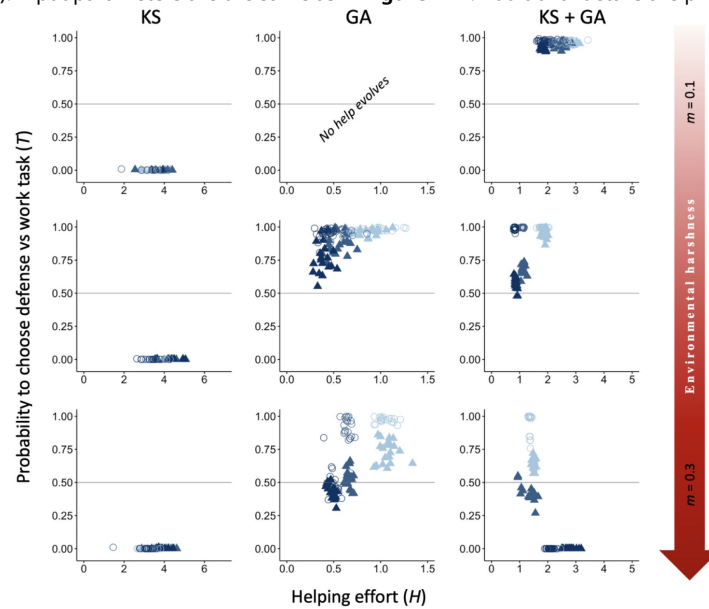


Table S1.

Supplementary data for Figure 2.

Mean values are shown for dispersal propensity, survival probability, group size ($\pm$ SD), number of floaters ($\pm$ SD) and within-group relatedness for three environmental qualities ranging from benign ($m=0.1$) to harsh ($m=0.3$) across 20 replicas. Selective forces at play include kin selection (KS), group augmentation (GA), or both (KS + GA). The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	KS						GA						KS + GA					
	$m=0.1$		$m=0.2$		$m=0.3$		$m=0.1$		$m=0.2$		$m=0.3$		$m=0.1$		$m=0.2$		$m=0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.95	0.92	0.94	0.90	0.91	0.84	0.01	0.01	0.11	0.10	0.37	0.33	0.07	0.07	0.22	0.22	0.43	0.52
Survival	0.74	0.74	0.65	0.65	0.57	0.57	0.90	0.90	0.79	0.79	0.66	0.67	0.89	0.89	0.77	0.77	0.65	0.65
Group size	1.16 ± 0.16	1.34 ± 0.30	1.14 ± 0.12	1.25 ± 0.19	1.14 ± 0.08	1.29 ± 0.11	9.85 ± 0.08	9.83 ± 0.05	6.94 ± 0.43	7.52 ± 0.41	3.04 ± 0.35	3.40 ± 0.36	16.42 ± 0.10	16.55 ± 0.07	6.42 ± 0.14	6.61 ± 0.11	2.95 ± 0.25	2.83 ± 0.05
Number of floaters	14280 ± 1352	15616 ± 2322	9778 ± 711	10352 ± 1082	6956 ± 330	7482 ± 395	297 ± 23	297 ± 22	3512 ± 104	3428 ± 101	5925 ± 137	5905 ± 122	5956 ± 121	5968 ± 107	7523 ± 252	7853 ± 201	7353 ± 1173	9741 ± 224
Relatedness	0.37	0.36	0.54	0.52	0.73	0.71	0	0	0	0	0	0	0.56	0.57	0.57	0.57	0.65	0.60

Table S2.

Supplementary data for the effect of increasing the baseline survival x_0 to favor the evolution of division of labor under only kin selection shown in Figure S1

Mean values are shown for dispersal propensity, survival probability, group size ($\pm$ SD), number of floaters ($\pm$ SD) and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Results are shown for $x_0 = 1.5$ (default), $x_0 = 3.5$ and $x_0 = 10$. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	$x_0 = 1.5$						$x_0 = 4.5$						$x_0 = 10$					
	$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.95	0.92	0.94	0.90	0.91	0.84	0.84	0.85	0.89	0.86	0.89	0.85	0.79	0.83	0.88	0.85	0.9	0.86
Survival	0.74	0.74	0.65	0.65	0.57	0.57	0.89	0.89	0.79	0.79	0.69	0.69	0.90	0.90	0.80	0.80	0.70	0.70
Group size	1.16 ± 0.16	1.34 ± 0.30	1.14 ± 0.12	1.25 ± 0.19	1.14 ± 0.08	1.29 ± 0.11	3.02 ± 0.33	3.14 ± 0.14	1.61 ± 0.26	1.86 ± 0.21	1.33 ± 0.13	1.50 ± 0.09	4.19 ± 0.19	3.75 ± 0.20	1.68 ± 0.22	1.96 ± 0.07	1.28 ± 0.11	1.48 ± 0.07
Number of floaters	14280 ± 1352	15616 ± 2322	9778 ± 711	10352 ± 1082	6956 ± 330	7482 ± 395	53241 ± 5074	61767 ± 1386	22911 ± 2167	25655 ± 1903	12672 ± 836	13848 ± 548	59534 ± 3345	65170 ± 2536	23585 ± 1799	27291 ± 560	12688 ± 714	14252 ± 618
Relatedness	0.37	0.36	0.54	0.52	0.73	0.71	0.15	0.15	0.28	0.28	0.45	0.44	0.17	0.15	0.27	0.27	0.44	0.43

Table S3.

Supplementary data for the effect of reducing the incentives to disperse to favor the evolution of division of labor under only kin selection shown in Figure S2

The increased incentive was achieved by reducing the parameter f that signifies the mean number of groups a floater samples for becoming a breeder from 2 (default) to 1. Mean values are shown for dispersal propensity, survival probability, group size ($\pm$ SD), number of floaters ($\pm$ SD) and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Results are shown for $x_0 = 1.5$ (default), $x_0 = 3.5$ and $x_0 = 10$. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	$x_0 = 1.5$						$x_0 = 4.5$						$x_0 = 10$					
	$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.82	0.75	0.79	0.79	NA	NA	0.23	0.24	0.49	0.50	0.72	0.68	0.25	0.25	0.44	0.47	0.49	0.66
Survival	0.74	0.74	0.65	0.65	0	0	0.89	0.89	0.79	0.79	0.69	0.69	0.90	0.90	0.80	0.80	0.70	0.70
Group size	1.90 ± 1.07	2.34 ± 1.26	1.71 ± 0.62	1.75 ± 0.75	0	0	13.09 ± 0.23	13.35 ± 0.40	4.84 ± 0.38	4.90 ± 0.08	2.04 ± 0.06	2.33 ± 0.05	14.29 ± 0.26	14.59 ± 0.16	5.17 ± 0.37	5.30 ± 0.08	3.10 ± 0.26	2.47 ± 0.05
Number of floaters	14116 ± 972	14468 ± 1058	9986 ± 773	9957 ± 859	0	0	18182 ± 894	19426 ± 1767	18440 ± 1678	19531 ± 337	13682 ± 154	14236 ± 185	21746 ± 639	22100 ± 361	16541 ± 1838	18838 ± 378	9839 ± 265	14274 ± 515
Relatedness	0.40	0.39	0.55	0.56	NA	NA	0.32	0.32	0.37	0.36	0.46	0.46	0.29	0.29	0.38	0.36	0.52	0.46

Table S4.

Supplementary data for effect of reducing within-group relatedness by half to mimic sexual reproduction shown in Figure S3

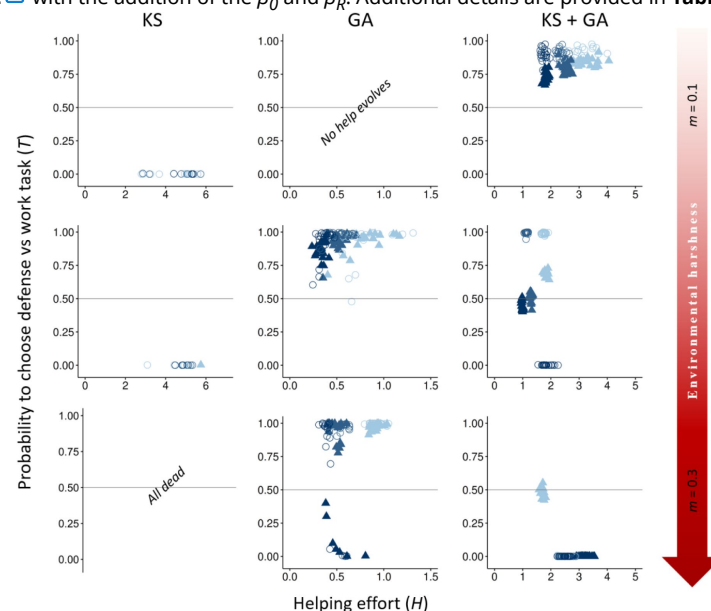
Mean values are shown for dispersal propensity, survival probability, group size ($\pm$ SD), number of floaters ($\pm$ SD) and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Selective forces at play include kin selection (KS), group augmentation (GA), or both (KS + GA). The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	KS						GA						KS + GA					
	$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.94	0.93	0.93	0.90	0.91	0.88	0.01	0.01	0.11	0.10	0.37	0.33	0.02	0.02	0.11	0.11	0.41	0.50
Survival	0.74	0.74	0.65	0.65	0.57	0.57	0.90	0.90	0.79	0.79	0.66	0.67	0.90	0.90	0.79	0.79	0.66	0.65
Group size	1.20	1.29	1.16	1.26	1.14	1.22	9.85	9.83	6.94	7.52	3.04	3.40	18.37	18.54	7.66	7.86	3.16	2.78
	± 0.19	± 0.27	± 0.12	± 0.18	± 0.08	± 0.14	± 0.08	± 0.05	± 0.43	± 0.41	± 0.35	± 0.36	± 0.10	± 0.09	± 0.15	± 0.21	± 0.53	± 0.09
Number of floaters	14491	15139	9820	10300	6869	7123	297	297	3512	3428	5925	5905	2216	2218	4164	4179	7327	9020
	± 1329	± 2030	± 682	± 1004	± 291	± 486	± 23	± 22	± 104	± 101	± 137	± 122	± 74	± 83	± 124	± 119	± 1378	± 346
Relatedness	0.18	0.20	0.27	0.26	0.36	0.36	0	0	0	0	0	0	0.30	0.31	0.31	0.31	0.32	0.30

Figure S4.

Effect of adding a reaction norm of dispersal and immigration propensity to dominance value.

The evolutionary equilibria for phenotypic levels of helping and task specialization are shown at three different levels of environmental quality, ranging from benign ($m=0.1$) to harsh ($m=0.3$), and for three different levels of cost of help on survival (light blue, $x_h=3$; blue, $x_h=5$; and dark blue, $x_h=7$), across 20 replicas. The vertical axis expresses the probability of individuals choosing a defensive task with a cost to survival versus a work task with a cost to their dominance rank. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$) or when no restrictions were introduced to the task performed by the group members ($\circ$). In each environment, additional details are given on the selective forces that play a role in the evolution of help and task specialization: help can only evolve by kin selection (KS), group augmentation (GA), or both (KS + GA). Input parameters are the same as in Figure 2 with the addition of the β_D and β_R . Additional details are provided in Table S5.



	KS						GA						KS + GA					
	<i>m</i> = 0.1		<i>m</i> = 0.2		<i>m</i> = 0.3		<i>m</i> = 0.1		<i>m</i> = 0.2		<i>m</i> = 0.3		<i>m</i> = 0.1		<i>m</i> = 0.2		<i>m</i> = 0.3	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	1.00	0.94	1.00	0.96	NA	NA	0.00	0.00	0.16	0.15	0.67	0.51	0.08	0.08	0.20	0.20	0.43	0.45
β_R	-3.71	-3.17	-4.27	-4.18	NA	NA	3.99	3.99	-0.30	-0.48	-4.61	-4.52	2.71	2.72	0.67	2.17	2.45	3.49
Survival	0.74	0.74	0.65	0.65	NA	NA	0.90	0.90	0.78	0.78	0.65	0.64	0.89	0.89	0.77	0.77	0.66	0.65
Group size	1.00 ±0.00	1.27 ±0.39	1.01 ±0.06	1.12 ±0.22	NA	NA	9.76 ±0.08	9.75 ±0.11	6.15 ±0.78	6.77 ±0.99	1.87 ±0.57	2.44 ±0.16	16.21 ±0.10	16.23 ±0.10	6.60 ±0.16	6.85 ±0.17	3.13 ±0.38	3.15 ±0.06
Number of floaters	13928 ±123	15486 ±2148	9477 ±236	9943 ±928	NA	NA	25 ±9	26 ±9	4630 ±493	4740 ±1303	7350 ±504	7367 ±730	6408 ±148	6537 ±144	7059 ±188	7190 ±177	8055 ±1161	8935 ±150
Relatedness	0.31	0.34	0.31	0.30	NA	NA	0	0	0	0	0	0	0.37	0.37	0.48	0.43	0.42	0.36

Table S5.

Supplementary data for the effect of adding a reaction norm of dispersal and immigration propensity to dominance value shown in Figure S4 [↗](#).

Mean values are shown for dispersal propensity, survival probability, group size ($\pm$ SD), number of floaters ($\pm$ SD) and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Selective forces at play include kin selection (KS), group augmentation (GA), or both (KS + GA). The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

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Reviewer #1 (Public review):

This paper presents a computational model of the evolution of two different kinds of helping ("work," presumably denoting provisioning, and defense tasks) in a model inspired by

cooperatively breeding vertebrates. The helpers in this model are a mix of previous offspring of the breeder and floaters that might have joined the group, and can either transition between the tasks as they age or not. The two types of help have differential costs: "work" reduces "dominance value," (DV), a measure of competitiveness for breeding spots, which otherwise goes up linearly with age, but defense reduces survival probability. Both eventually might preclude the helper from becoming a breeder and reproducing. How much the helpers help, and which tasks (and whether they transition or not), as well as their propensity to disperse, are all evolving quantities. The authors consider three main scenarios: one where relatedness emerges from the model, but there is no benefit to living in groups, one where there is no relatedness, but living in larger groups gives a survival benefit (group augmentation, GA), and one where both effects operate. The main claim is that evolving defensive help or division of labor requires the group augmentation; it doesn't evolve through kin selection alone in the authors' simulations.

This is an interesting model, and there is much to like about the complexity that is built in. Individual-based simulations like this can be a valuable tool to explore the complex interaction of life history and social traits. Yet, models like this also have to take care of both being very clear on their construction and exploring how some of the ancillary but potentially consequential assumptions affect the results, including robust exploration of the parameter space. I think the current manuscript falls short in these areas, and therefore, I am not yet convinced of the results.

In this round, the authors provided some clarity, but some questions still remain, and I remain unconvinced by a main assumption that was not addressed.

Based on the authors' response, if I understand the life history correctly, dispersers either immediately join another group (with 1-the probability of dispersing), or remain floaters until they successfully compete for a breeder spot or die? Is that correct? I honestly cannot decide because this seems implicit in the first response but the response to my second point raises the possibility of not working while floating but can work if they later join a group as a subordinate. If it is the case that floaters can have multiple opportunities to join groups as subordinates (not as breeders; I assume that this is the case for breeding competition), this should be stated, and more details about how.

So there is still some clarification to be done, and more to the point, the clarification that happened only happened in the response. The authors should add these details to the main text. Currently, the main text only says vaguely that joining a group after dispersing "is also controlled by the same genetic dispersal predisposition" without saying how.

In response to my query about the reasonableness of the assumption that floaters are in better condition (in the KS treatment) because they don't do any work, the authors have done some additional modeling but I fail to see how that addresses my point. The additional simulations do not touch the feature I was commenting on, and arguably make it stronger (since assuming a positive β_r -which btw is listed as 0 in Table 1- would make floaters on average be even more stronger than subordinates). It also again confuses me with regard to the previous point, since it implies that now dispersal is also potentially a lifetime event. Is that true?

Meanwhile, the simplest and most convincing robustness check, which I had suggested last round, is not done: simply reduce the increase in the R of the floater by age relative to subordinates. I suspect this will actually change the results. It seems fairly transparent to me that an average floater in the KS scenario will have R about 15-20% higher than the subordinates (given no defense evolves, $y_h=0.1$ and H_{work} evolves to be around 5, and the average lifespan for both floaters and subordinates are in the range of 3.7-2.5 roughly, depending on m). That could be a substantial advantage in competition for breeding spots,

depending on how that scramble competition actually works. I asked about this function in the last round (how non-linear is it?) but the authors seem to have neglected to answer.

More generally, I find that the assumption (and it is an assumption) floaters are better off than subordinates in a territory to be still questionable. There is no attempt to justify this with any data, and any data I can find points the other way (though typically they compare breeders and floaters, e.g.: <https://bioone.org/journals/ardeola/volume-63/issue-1/arla.63.1.2016.rp3/The-Unknown-Life-of-Floaters--The-Hidden-Face-of/10.13157/arla.63.1.2016.rp3.full> concludes "the current preliminary consensus is that floaters are 'making the best of a bad job'."). I think if the authors really want to assume that floaters have higher dominance than subordinates, they should justify it. This is driving at least one and possibly most of the key results, since it affects the reproductive value of subordinates (and therefore the costs of helping).

Regarding division of labor, I think I was not clear so will try again. The authors assume that the group reproduction is $1 + H_{\text{total}} / (1 + H_{\text{total}})$, where H_{total} is the sum of all the defense and work help, but with the proviso that if one of the totals is higher than " H_{max} ", the average of the two totals (plus k_m , but that's set to a low value, so we can ignore it), it is replaced by that. That means, for example, if total "work" help is 10 and "defense" help is 0, total help is given by 5 (well, 5.1 but will ignore k_m). That's what I meant by "marginal benefit of help is only reduced by a half" last round, since in this scenario, adding 1 to work help would make total help go to 5.5 vs. adding 1 to defense help which would make it go to 6. That is a pretty weak form of modeling "both types of tasks are necessary to successfully produce offspring" as the newly added passage says (which I agree with), since if you were getting no defense by a lot of food, adding more food should plausibly have no effect on your production whatsoever (not just half of adding a little defense). This probably explains why often the "division of labor" condition isn't that different than the no DoL condition.

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Reviewer #2 (Public review):

Summary:

This paper formulates an individual-based model to understand the evolution of division of labor in vertebrates. The model considers a population subdivided in groups, each group has a single asexually-reproducing breeder, other group members (subordinates) can perform two types of tasks called "work" or "defense", individuals have different ages, individuals can disperse between groups, each individual has a dominance rank that increases with age, and upon death of the breeder a new breeder is chosen among group members depending on their dominance. "Workers" pay a reproduction cost by having their dominance decreased, and "defenders" pay a survival cost. Every group member receives a survival benefit with increasing group size. There are 6 genetic traits, each controlled by a single locus, that control propensities to help and disperse, and how task choice and dispersal relate to dominance. To study the effect of group augmentation without kin selection, the authors cross-foster individuals to eliminate relatedness. The paper allows for the evolution of the 6 genetic traits under some different parameter values to study the conditions under which division of labour evolves, defined as the occurrence of different subordinates performing "work" and "defense" tasks. The authors envision the model as one of vertebrate division of labor.

The main conclusion of the paper is that group augmentation is the primary factor causing the evolution of vertebrate division of labor, rather than kin selection. This conclusion is drawn because, for the parameter values considered, when the benefit of group augmentation is set to zero, no division of labor evolves and all subordinates perform "work" tasks but no "defense" tasks.

Strengths:

The model incorporates various biologically realistic details, including the possibility to evolve age polytheism where individuals switch from "work" to "defence" tasks as they age or vice versa, as well as the possibility of comparing the action of group augmentation alone with that of kin selection alone.

Weaknesses:

The model and its analysis is limited, which makes the results insufficient to reach the main conclusion that group augmentation and not kin selection is the primary cause of the evolution of vertebrate division of labor. There are several reasons.

First, the model strongly restricts the possibility that kin selection is relevant. The two tasks considered essentially differ only by whether they are costly for reproduction or survival. "Work" tasks are those costly for reproduction and "defense" tasks are those costly for survival. The two tasks provide the same benefits for reproduction (eqs. 4, 5) and survival (through group augmentation, eq. 3.1). So, whether one, the other, or both tasks evolve presumably only depends on which task is less costly, not really on which benefits it provides. As the two tasks give the same benefits, there is no possibility that the two tasks act synergistically, where performing one task increases a benefit (e.g., increasing someone's survival) that is going to be compounded by someone else performing the other task (e.g., increasing that someone's reproduction). So, there is very little scope for kin selection to cause the evolution of labour in this model. Note synergy between tasks is not something unusual in division of labour models, but is in fact a basic element in them, so excluding it from the start in the model and then making general claims about division of labour is unwarranted. I made this same point in my first review, although phrased differently, but it was left unaddressed.

Second, the parameter space is very little explored. This is generally an issue when trying to make general claims from an individual-based model where only a very narrow parameter region has been explored of a necessarily particular model. However, in this paper, the issue is more evident. As in this model the two tasks ultimately only differ by their costs, the parameter values specifying their costs should be varied to determine their effects. Instead, the model sets a very low survival cost for work ($y_h=0.1$) and a very high survival cost for defense ($x_h=3$), the latter of which can be compensated by the benefit of group augmentation ($x_n=3$). Some very limited variation of x_h and x_n is explored, always for very high values, effectively making defense unevolvable except if there is group augmentation. Hence, as I stated in my previous review, a more extensive parameter exploration addressing this should be included, but this has not been done. Consequently, the main conclusion that "division of labor" needs group augmentation is essentially enforced by the limited parameter exploration, in addition to the first reason above.

Third, what is called "division of labor" here is an overinterpretation. When the two tasks evolve, what exists in the model is some individuals that do reproduction-costly tasks (so-called "work") and survival-costly tasks (so-called "defense"). However, there are really no two tasks that are being completed, in the sense that completing both tasks (e.g., work and defense) is not necessary to achieve a goal (e.g., reproduction). In this model there is only one task (reproduction, equation 4,5) to which both "tasks" contribute equally and so one task doesn't need to be completed if the other task compensates for it. So, this model does not actually consider division of labor.

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Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

This paper presents a computational model of the evolution of two different kinds of helping ("work," presumably denoting provisioning, and defense tasks) in a model inspired by cooperatively breeding vertebrates. The helpers in this model are a mix of previous offspring of the breeder and floaters that might have joined the group, and can either transition between the tasks as they age or not. The two types of help have differential costs: "work" reduces "dominance value," (DV), a measure of competitiveness for breeding spots, which otherwise goes up linearly with age, but defense reduces survival probability. Both eventually might preclude the helper from becoming a breeder and reproducing. How much the helpers help, and which tasks (and whether they transition or not), as well as their propensity to disperse, are all evolving quantities. The authors consider three main scenarios: one where relatedness emerges from the model, but there is no benefit to living in groups, one where there is no relatedness, but living in larger groups gives a survival benefit (group augmentation, GA), and one where both effects operate. The main claim is that evolving defensive help or division of labor requires the group augmentation; it doesn't evolve through kin selection alone in the authors' simulations.

This is an interesting model, and there is much to like about the complexity that is built in. Individual-based simulations like this can be a valuable tool to explore the complex interaction of life history and social traits. Yet, models like this also have to take care of both being very clear on their construction and exploring how some of the ancillary but potentially consequential assumptions affect the results, including robust exploration of the parameter space. I think the current manuscript falls short in these areas, and therefore, I am not yet convinced of the results. Much of this is a matter of clearer and more complete writing: the Materials and Methods section in particular is incomplete or vague in some important junctions. However, there are also some issues with the assumptions that are described clearly.

Below, I describe my main issues, mostly having to do with model features that are unclear, poorly motivated (as they stand), or potentially unrealistic or underexplored.

We would like to thank the reviewer for the thoughtful comments that helped us to greatly improve the clarity of our paper.

One of the main issues I have is that there is almost no information on what happens to dispersers in the model. Line 369-67 states dispersers might join another group or remain as floaters, but gives no further information on how this is determined. Poring through the notation table also comes up empty as there is no apparent parameter affecting this consequential life history event. At some point, I convinced myself that dispersers remain floaters until they die or become breeders, but several points in the text contradict this directly (e.g., l 107). Clearly this is a hugely important model feature since it determines fitness cost and benefits of dispersal and group size (which also affects relatedness and/or fitness depending on the model). There just isn't enough information to understand this crucial component of the model, and without it, it is hard to make sense of the model output.

We use the same dispersal gene β to represent the likelihood an individual will either leave or join a group, thereby quantifying both dispersal and immigration using the same parameter. Specifically, individuals with higher β are more likely to remain as floaters (i.e., disperse from their natal group to become a breeder elsewhere), whereas those with lower β are either more likely to remain in their natal group as subordinates (i.e., queue in a group for the breeding position) or join another group if they dispersed.

We added in the text “Dispersers may migrate to another group to become subordinates or remain as floaters waiting for breeding opportunities, which is also controlled by the same genetic dispersal propensity as subordinates” to clarify this issue. We also added in Table 1 that β is the “genetic predisposition to disperse versus remain in a group”, and to Figure 1 that “subordinates in the group (natal and immigrants) [...]” after we already clarified that “Dispersers/floaters may join a random group to become subordinates.”

Related to that, it seems to be implied (but never stated explicitly) that floaters do not work, and therefore their DV increases linearly with age (H_{work} in eq.2 is zero). That means any floaters that manage to stick around long enough would have higher success in competition for breeding spots relative to existing group members. How realistic is this? I think this might be driving the kin selection-only results that defense doesn't evolve without group augmentation (one of the two main ways). Any subordinates (which are mainly zero in the no GA, according to the SI tables; this assumes $N = \text{breeder} + \text{subordinates}$, but this isn't explicit anywhere) would be outcompeted by floaters after a short time (since they evolve high H and floaters don't), which in turn increases the benefit of dispersal, explaining why it is so high. Is this parameter regime reasonable? My understanding is that floaters often aren't usually high resource holding potential individuals (either b/c high RHP ones would get selected out of the floater population by establishing territories or b/c floating isn't typically a thriving strategy, given that many resources are tied to territories). In this case, the assumption seems to bias things towards the floaters and against subordinates to inherit territories. This should be explored either with a higher mortality rate for floaters and/or a lower DV increase, or both.

When it comes to floaters replacing dead breeders, the authors say a bit more, but again, the actual equation for the scramble competition (which only appears as "scramble context" in the notation table) is not given. Is it simply proportional to $R_i / \sum_j R_j$? Or is there some other function used? What are the actual numbers of floaters per breeding territory that emerge under different parameter values? These are all very important quantities that have to be described clearly.

Although it is true that dispersers do not work when they are floaters, they may later help if they immigrate into a group as a subordinate. Consequently, immigrant subordinates have no inherent competitive advantage over natal subordinates (as step 2.2. “Join a group” is followed by step 3. “Help”, which occurs before step 5. “Become a breeder”). Nevertheless, floaters can potentially outcompete subordinates of the same age if they attempt to breed without first queuing as a subordinate (step 5) when subordinates are engaged in work tasks. We believe that this assumption is realistic and constitutes part of the costs associated with work tasks. However, floaters are at a disadvantage for becoming a breeder because: (1) floaters incur higher mortality than individuals within groups (Eq. 3); and (2) floaters may only attempt to become breeders in some breeding cycles (versus subordinate group members, who are automatically candidates for an open breeding position in the group in each cycle). Therefore, due to their higher mortality, floaters are rarely older than individuals within groups, which heavily influences their dominance value and competitiveness. Additionally, any competitive advantage that floaters might have over other subordinate group members is unlikely to drive the kin selection-only results because subordinates would

preferably choose defense tasks instead of work tasks so as not to be at a competitive disadvantage compared to floaters.

Regarding whether floaters aren't usually high resource holding potential (RHP) individuals and, therefore, our assumptions might be unrealistic; empirical work in a number of species has shown that dispersers are not necessarily those of lower RHP or of lower quality. In fact, according to the ecological constraints hypothesis, one might predict that high quality individuals are the ones that disperse because only individuals in good condition (e.g., larger body size, better energy reserves) can afford the costs associated with dispersal (Cote et al., 2022). To allow differences in dispersal propensity depending on RHP, we extended our model in the Supplemental Materials by incorporating a reaction norm of dispersal based on their rank ($D = 1 / (1 + \exp(\beta_R * R - \beta_0))$) under the section "Dominance-dependent dispersal propensities" and now referenced in L195. This approach allows individuals to adjust their dispersal strategy to their competitiveness and to avoid kin competition by remaining as a subordinate in another group. Results show that the addition of the reaction norm of dispersal to rank did not qualitatively influence the results described in the main text.

We also added "number of floaters" present in the whole population to the summary tables as requested.

As a side note, the "scramble context" we mention was an additional implementation in which we made rank independent of age. However, since the main conclusions remained unchanged, we decided to remove it for simplicity from the final manuscript, but we forgot to remove it from Table 1 before submission.

I also think the asexual reproduction with small mutations assumption is a fairly strong one that also seems to bias the model outcomes in a particular way. I appreciate that the authors actually measured relatedness within groups (though if most groups under KS have no subordinates, that relatedness becomes a bit moot), and also eliminated it with their ingenious swapping-out-subordinates procedure. The fact remains that unless they eliminate relatedness completely, average relatedness, by design, will be very high. (Again, this is also affected by how the fate of the dispersers is determined, but clearly there isn't a lot of joining happening, just judging from mean group sizes under KS only.) This is, of course, why there is so much helping evolving (even if it's not defensive) unless they completely cut out relatedness.

As we showed in the Supplementary Tables and the section on relatedness in the SI ("Kin selection and the evolution of division of labor"), high relatedness does not appear to explain our results. In evolutionary biology generally and in game theory specifically (with the exception of models on sexual selection or sex-specific traits), asexual reproduction is often modelled because it reduces unnecessary complexity. To further study the effect of relatedness on kin structures more closely resembling those of vertebrates, however, we created an additional "relatedness structure level", where we shuffled half of the philopatric offspring using the same method used to remove relatedness completely, effectively reducing within-group relatedness structure by half. As shown in the new Figure S3, the conclusions of the model remain unchanged.

Finally, the "need for division of labor" section is also unclear, and its construction also would seem to bias things against division of labor evolving. For starters, I don't understand the rationale for the convoluted way the authors create an incentive for division of labor. Why not implement something much simpler, like a law of minimum (i.e., the total effect of helping is whatever the help amount for the lowest value task is) or more intuitively: the fecundity is simply a function of "work" help (draw Poisson number of offspring) and survival of offspring (draw binomial from the fecundity) is a function of the "defense" help. As it is, even though the authors say they require division of labor, in

fact, they only make a single type of help marginally less beneficial (basically by half) if it is done more than the other. That's a fairly weak selection for division of labor, and to me it seems hard to justify. I suspect either of the alternative assumptions above would actually impose enough selection to make division of labor evolve even without group augmentation.

In nature, multiple tasks are often necessary to successfully rear offspring. We simplify this principle in the model by maximizing reproductive output when both tasks are carried out to a similar extent, allowing for some flexibility from the mean. We added to the manuscript “For example, in many cooperatively breeding birds, the primary reasons that individuals fail to produce offspring are (1) starvation, which is mitigated by the feeding of offspring, and (2) nest depredation, which is countered by defensive behavior. Consequently, both types of tasks are necessary to successfully produce offspring, and focusing solely on one while neglecting the other is likely to result in lower reproductive success than if both tasks are performed by individuals within the group.”

Regarding making fecundity a function of work tasks and offspring survival as a function of defensive tasks, these are actually equivalent in model terms, as it's the same whether breeders produce three offspring and two die, or if they only produce one. This represents, of course, an oversimplification of the natural context, where breeding unsuccessfully is more costly (in terms of time and energy investment) than not breeding at all.

Overall, this is an interesting model, but the simulation is not adequately described or explored to have confidence in the main conclusions yet. Better exposition and more exploration of alternative assumptions and parameter space are needed.

We hope that our clarifications and extension of the model satisfy your concerns.

Reviewer #2 (Public review):

Summary:

This paper formulates an individual-based model to understand the evolution of division of labor in vertebrates. A main conclusion of the paper is that direct fitness benefits are the primary factor causing the evolution of vertebrate division of labor, rather than indirect fitness benefits.

Strengths:

The paper formulates an individual-based model that is inspired by vertebrate life history. The model incorporates numerous biologically realistic details, including the possibility to evolve age polytheism where individuals switch from work to defence tasks as they age or vice versa, as well as the possibility of comparing the action of group augmentation alone with that of kin selection alone.

Weaknesses:

The model makes assumptions that restrict the possibility that kin selection leads to the evolution of helping. In particular, the model assumes that in the absence of group augmentation, subordinates can only help breeders but cannot help non-breeders or increase the survival of breeders, whereas with group augmentation, subordinates can help both breeders and non-breeders and increase the survival of breeders. This is unrealistic as subordinates in real organisms can help other subordinates and increase the survival of non-breeders, even in the absence of group augmentation, for instance, with targeted helping to dominants or allies. This restriction artificially limits the ability of kin selection alone to lead to the evolution of helping, and potentially to division of

labor. Hence, the conclusion that group augmentation is the primary driving factor driving vertebrate division of labor appears forced by the imposed restrictions on kin selection. The model used is also quite particular, and so the claimed generality across vertebrates is not warranted.

We would like to thank the reviewer for the in-depth review. We respond to these and other comments below.

I describe some suggestions for improving the paper below, more or less in the paper's order.

First, the introduction goes to great lengths trying to convince the reader that this model is the first in this or another way, particularly in being only for vertebrates, as illustrated in the abstract where it is stated that "we lack a theoretical framework to explore the conditions under which division of labor is likely to evolve" (line 13). However, this is a risky and unnecessary motivation. There are many models of division of labor and some of them are likely to be abstract enough to apply to vertebrates even if they are not tailored to vertebrates, so the claims for being first are not only likely to be wrong but will put many readers in an antagonistic position right from the start, which will make it harder to communicate the results. Instead of claiming to be the first or that there is a lack of theoretical frameworks for vertebrate division of labor, I think it is enough and sufficiently interesting to say that the paper formulates an individual-based model motivated by the life history of vertebrates to understand the evolution of vertebrate division of labor. You could then describe the life history properties that the model incorporates (subordinates can become reproductive, low relatedness, age polyethism, etc.) without saying this has never been done or that it is exclusive to vertebrates; indeed, the paper states that these features do not occur in eusocial insects, which is surprising as some "primitively" eusocial insects show them. So, in short, I think the introduction should be extensively revised to avoid claims of being the first and to make it focused on the question being addressed and how it is addressed. I think this could be done in 2-3 paragraphs without the rather extensive review of the literature in the current introduction.

We have revised the novelty statements in the Introduction by more clearly emphasizing how our model addresses gaps in the existing literature. More details are provided in the comments below.

Second, the description of the model and results should be clarified substantially. I will give specific suggestions later, but for now, I will just say that it is unclear what the figures show. First, it is unclear what the axes in Figure 2 show, particularly for the vertical one. According to the text in the figure axis, it presumably refers to T , but T is a function of age t , so it is unclear what is being plotted. The legend explaining the triangle and circle symbols is unintelligible (lines 227-230), so again it is unclear what is being plotted; part of the reason for this unintelligibility is that the procedure that presumably underlies it (section starting on line 493) is poorly explained and not understandable (I detail why below). Second, the axes in Figure 3 are similarly unclear. The text in the vertical axis in panel A suggests this is T , however, T is a function of t and γ_t , so something else must be being done to plot this. Similarly, in panel B, the horizontal axis is presumably R , but R is a function of t and of the helping genotype, so again some explanation is lacking. In all figures, the symbol of what is being plotted should be included.

We added the symbols of the variables to the Figure axes to increase clarity. In Figure 3A, we corrected the subindex t in the x-axis; it should be subindex R (reaction norm to dominance

rank instead of age). As described in Table 1, all values of T , H and R are phenotypically expressed values. For instance, T values are the phenotypically expressed values from the individuals in the population according to their genetic gamma values and their current dominance rank at a given time point.

Third, the conclusions sound stronger than the results are. A main conclusion of the paper is that "kin selection alone is unlikely to select for the evolution of defensive tasks and division of labor in vertebrates" (lines 194-195). This conclusion is drawn from the left column in Figure 2, where only kin selection is at play, and the helping that evolves only involves work rather than defense tasks. This conclusion follows because the model assumes that without group augmentation (i.e., $x_n=0$, the kin selection scenario), subordinates can only help breeders to reproduce but cannot help breeders or other subordinates to survive, so the only form of help that evolves is the least costly, not the most beneficial as there is no difference in the benefits given among forms of helping. This assumption is unrealistic, particularly for vertebrates where subordinates can help other group members survive even in the absence of group augmentation (e.g., with targeted help to certain group members, because of dominance hierarchies where the helping would go to the breeder, or because of alliances where the helping would go to other subordinates). I go into further details below, but in short, the model forces a narrow scope for the kin selection scenario, and then the paper concludes that kin selection alone is unlikely to be of relevance for the evolution of vertebrate division of labor. This conclusion is particular to the model used, and it is misleading to suggest that this is a general feature of such a particular model.

The scope of this paper was to study division of labor in cooperatively breeding species with fertile workers (i.e., primarily vertebrates), in which help is exclusively directed towards breeders to enhance offspring production (i.e., alloparental care). Our focus is in line with previous work in most other social animals, including eusocial insects and humans, which emphasizes how division of labor maximizes group productivity. Other forms of "general" help are not considered in the paper, and such forms of help are rarely considered in cooperatively breeding vertebrates or in the division of labor literature, as they do not result in task partitioning to enhance productivity.

Overall, I think the paper should be revised extensively to clarify its aims, model, results, and scope of its conclusions.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

I reserved this section for more minor comments, relating to clarity and a general admonition to give us more detail and exploration of some basic population genetic quantities.

Another minor point, although depending on whether I assume right or wrong, it could be major: I am not entirely sure that dispersers help in the groups they join as helpers, because of line 399, which states specifically that individuals who do remain in natal territories do. But I assume dispersers help (elsewhere, the authors state helping is not conditional on relatedness to the breeder). Otherwise, this model becomes even weirder for me. Either way, please clarify.

Apologies if this was not clear. Immigrants that join a group (so dispersers from another group) as a subordinate help and queue for a breeding position, as does any natal subordinate born into the group. We rephased the sentence to "Subordinate group members, either natal or immigrants to the group, [...]"

More generally, in simulation studies like this, there can be interactions between the strength of selection (which affects overall genetic variation maintained in the population), population size, and mutation rate/size, which can affect, for example, relatedness values. None of these quantities is explored here (and their interactions are not quantified), so it is not possible to evaluate the robustness of any of these results.

Thank you for your comments about the parameter landscape. It is important to point out that variations in the mutation rate do not qualitatively affect our results, as this is something we explored in previous versions of the model (not shown). Briefly, we find that variations in the mutation rates only alter the time required to reach equilibrium. Increasing the step size of mutation diminishes the strength of selection by adding stochasticity and reducing the genetic correlation between offspring and their parents. Population size could, in theory, affect our results, as small populations are more prone to extinction. Since this was not something we planned to explore in the paper directly, we specifically chose a large population size, or better said, a large number of territories (i.e. 5000) that can potentially host a large population.

The authors also never say how it is actually determined. There is the evolved helping variable, and there is also the evolved reaction norm. I assume that the actual amount of help of each type is given by the product of T (equation 1) and H (for defense) and $(1-T)$ and H (for work), but this should be stated explicitly.

Help provided is an interaction between H (total effort) and T (proportion of total effort invested in each type of task). To clarify the distinction between these two processes, we have now added “Hence, the gene α regulates the amount of help expressed, while the genes γ determine which specific helping tasks are performed at different time points in the breeding cycle”.

It is also weird that after introducing the T variable as a function of age, Figure 3 actually depicts it as a function of dominance value.

Thank you for pointing out an error in Eq. 1. This inequality was indeed written incorrectly in the paper (but is correct in the model code); it is dominance rank instead of age (see code in Individual.cpp lines 99-119). We corrected this mistake throughout the manuscript.

What is “scramble context”?

“Scramble context” was an additional implementation that we decided to remove from the final manuscript, but we forgot to remove from Table 1 before submission. We have now removed it from the table.

Reviewer #2 (Recommendations for the authors):

Some specific comments:

(1) L 31: “All theoretical...” These absolute statements are risky and unnecessary.

Rephrased to “To date, most theoretical and empirical work...”

(2) L 46: I believe Tom Wenseleers has published on the evolution of division of labor with reproductive workers and high within-colony conflict.

Tom Wenseleers has indeed produced some models on the evolution of cooperation in social insects where some workers may reproduce. However, these models focus on the relevance

of relatedness and policing selecting for a reduction in within-group conflict and the evolution of reproductive division of labor. Our model focuses instead on division of labor among workers (helpers). We have rephased this section to “task specialization is linked to sterility and where conflict of interest is generally low” to account for species of social insect in which variation in relatedness between group members and higher levels of reproductive conflict may arise. We also cited one of his papers.

(3) L 57: *Again, unnecessary categorical statements.*

Rephrased to “Although a great deal of recent empirical work highlights the importance of direct benefits in the evolution of cooperative breeding behavior in vertebrates [21–24], we lack understanding on the joint influence of direct and indirect fitness benefits in the evolution of division of labor.”

(4) L 67: *This is said to be a key distinction, but in the paper, such a key role is not clearly shown. This and other tangential points are unnecessary to keep the introduction to the point.*

The different fitness costs of different tasks is the basis of our model on division of labor. Therefore, this is a key distinction and basis from which to describe different tasks in the model. We have left this sentence unchanged.

(5) L 61-73: *"In vertebrates, however, helpers may obtain fitness benefits directly via reproduction..." Some social insects may do so as well. It seems unnecessary and incorrect to say that vertebrate sociality is fundamentally different from invertebrate one. I think it is sufficiently interesting to say this work aims to understand vertebrate division of labor, by explicitly modeling aspects of its life history, without saying this can't happen in invertebrates or that no other model has ever done anything like it.*

Our point is not that, in some social insects, workers cannot obtain direct fitness benefits, but that previous models where the focus is on the colony reproductive outcome are only a good approximation to eusocial insect with sterile workers. However, to make this clearer we have added “In vertebrates and social insect with fertile workers, however, helpers may obtain fitness benefits directly via [...]”.

(6) L 74-86: *By this point, the introduction reads like a series of disconnected comments without a clear point.*

In L60 we added: “Understanding how direct and indirect benefits interact is particularly important in systems where individuals may differentially bear the fitness costs of cooperation”. By adding this sentence, we emphasize our focus on the largely unexplored direct fitness benefits and costs, as well as their interaction with indirect fitness. We then proceed to explain why it is crucial to consider that tasks have varying direct fitness costs and how the fitness benefits derived from cooperation change with age and resource-holding potential. These elements are essential for studying the division of labour in species with totipotent workers.

(7) L 87: *This sentence gives a clear aim. It would be clearer if the introduction focused on this aim.*

With the new sentence added in L60 (see previous comment), we bring the focus to the main question that we are trying to address in this paper earlier in the Introduction.

(8) L 88: *"stochastic model" should be changed to "individual-based model".*

Done.

| (9) L 104: "limited number" is unclear. Say a fixed finite number, or something specific.

Done.

| (10) L 105: "unspecified number" is unclear. Say the number of subordinates emerges from the population dynamics.

Changed to "variable number of subordinate helpers, the number of which is shaped by population dynamics, with all group members capable of reproducing during their lifetime".

| (11) L 112: "Dispersers" is used, but in the previous lines 107-109, the three categories introduced used different terms. Those three terms introduced should be used consistently throughout the paper, without using two or more terms for one thing.

We use the term "disperser" to describe individuals that disperse from their natal group.

Dispersers can assume one of three roles: (1) they can join another group as "subordinates"; (2) they can join another group as "breeders" if they successfully outcompete others; or (3) they can remain as "floaters" if they fail to join a group. "Floaters" are individuals who persist in a transient state without access to a breeding territory, waiting for opportunities to join a group in an established territory. We rephased the sentence to "Dispersers cannot reproduce without acquiring a territory (denoted here as floaters)". This was also clarified in other instances where the term "dispersers" was used (e.g. L407). Other instances where this might not have been so clear, we replace "dispersers" with "floaters".

| (12) L 112: "(floaters)" Unclear parenthesis.

See previous comment.

| (13) L 115: There should be a reference to Methods around here.

Added a reference to Figure 1.

| (14) L 117: To be clearer, say instead that dominance value is a linearly increasing function of age as a proxy of RHP and a linearly decreasing function of help provided due to the costs of working tasks. And refer to equation 2.

Rephrased to "We use the term dominance value to designate the competitiveness of an individual compared to other candidates in becoming a breeder, regardless of group membership, that increases as a function of age, serving as a proxy for resource holding potential (RHP), and decreases as a function of help provided, reflecting costs to body condition from performing working tasks (Eq. 2)." We did not include "linearly" to keep it simpler, since it is clear from Eq. 2, which is now referenced here.

| (15) L 119: "Subordinate helpers". As all subordinates are helpers, the helper qualifier is confusing.

Subordinates are not necessarily helpers, as they can evolve help values of 0, hence, why we make it explicit here.

| (16) L 119: "choose". This terminology may be misleading. The way things are implemented in the model is that individuals are assigned a task depending on their

genetic traits gamma. Perhaps it would be better to use a less intentional term, like perform one of two tasks.

We changed “choose between two” to “engage in one of two”, which has less connotations of intentionality.

(17) L 124: "Subordinates can [...] exhibit task specialization that [...] varies with their dominance value". It should be that it varies with age.

Apologies. The equation was wrong; it does vary with dominance value. We corrected it accordingly.

(18) L 133: "maximised" This is apparently important for the modelling procedure, but it is completely unclear what it means. Equation 4 comes out of nowhere, and it is said that such an equation is the maximum amount of help that can affect fecundity. Why? What does this mean? If there is something that is maximised, this should be proven. This value is then used for something (line 507), but it is unclear why or what it is used for (it says "we use the value of Hmax instead" without saying what for, no justification for the listed inequalities are given, and the claimed maximisation of an unspecified variable at those H values is not proven). Moreover, the notation in this section is also unclear: what are the sums over? Also, Hdefence and Hwork should vary over the index that is summed over, but the notation suggests that those quantities don't vary.

We changed “maximized” to “greatest”, and we added a clarification to the rationality behind the maximization of the impact of help in the breeder’s productivity: “For example, in many cooperatively breeding birds, the primary reasons that breeders fail to produce offspring are (1) starvation, which is mitigated by the feeding of offspring, here considered as a work task, and (2) nest depredation, which is countered by defensive behavior. Consequently, both types of tasks are often necessary for successful reproduction, and focusing solely on one while neglecting the other is likely to result in lower reproductive success than if both tasks are performed by helpers within the group.”

We now also clarify that the sums are for help given within a group (L 507), and added indexes to the equations.

(19) L 152: "habitat saturation" How is this implemented? How is density dependence implemented? Or can the population size keep increasing indefinitely? It would be good to plot the population size over time, the group size over time, and the variance in group size over time. This could substantiate later statements about enhancing group productivity and could all be shown in the SI.

Habitat saturation emerges from population dynamics due to the limited availability of territories and the fluctuating number of individuals, leading highly productive environments to experience habitat saturation. Although the number of group members is not restricted in our model, the population could theoretically increase indefinitely. However, this is not observed in the results presented here, as we selected parameter landscapes that stabilize population numbers. We confined our parameters to those where the population neither increased indefinitely (nor collapsed), as we did not incorporate density-dependent mortality traits for simplification. Consequently, the group size in the SI, where the standard deviation is already included, closely represents group size at any other given time during equilibrium.

L 336: we changed “environments with habitat saturation” to “environments *that lead to* habitat saturation”, to increase clarity.

(20) L 152: *"lifecycle". Rather than the lifecycle, the figure describes the cycle of events in a single time step. The lifecycle (birth to death) goes over multiple time steps (as individuals live over multiple steps). So this figure shouldn't be called a life cycle.*

We changed "lifecycle" to "breeding cycle".

(21) L 156: *"generation". This is not a generation but a time step.*

We changed "generation" to "breeding cycle".

(22) L 157: *"previous life cycle" would mean that the productivity of a breeder depends on the number of helpers that its parents had, which is not what is meant.*

We changed "lifecycle" to "breeding cycle".

(23) L 158: *"Maximum productivity is achieved when different helping tasks are performed to a similar extent." Again, unclear why that is the case.*

We added a clarification on this, see response to comment 18.

(24) L 160: *"Dispersers/floaters". Use just one term for a single thing.*

See response to comment 11.

(25) L 162: *"dispersal costs". I don't recall these being described in Methods.*

Individuals that disperse do not enjoy the protection of living in a territory and within a group of other individuals, so they have a higher mortality risk, described in Eq. 3.3. (negative values in the exponential part of the equation increase survival). The cost of dispersal is the same as individuals that remain as floaters at a given time step.

(26) L 164: *"generation" -> time step.*

We changed this to "breeding cycle".

(27) L 170: *"Our results show that division of labor initially emerges because of direct fitness benefits..." This is a general statement, but the results are only particular to the model. So this statement and others in the manuscript should be particular to the model. Also, Figure 2 doesn't say anything about what evolves "initially" as it only plots evolutionary equilibria.*

We rephrased this statement to "Our results suggest that voluntary division of labor involving tasks with different fitness costs is more likely to emerge initially because of direct fitness benefits", to more accurately represent the conditions under which we modeled the division of labor.

Our reference to "initially" is regarding group formation (family groups versus aggregations of unrelated individuals or a mix). This is shown in the comparison between the different graphs at equilibrium. The initial state of the simulation is that all individuals disperse and do not cooperate.

(28) L 171: *"but a combination of direct and indirect fitness benefits leads to higher rates and more stable forms of division of labor". What do you mean by "higher rates and*

more stable forms of division of labor"? Say how division of labor is shown in the figure (with intermediate T?).

Yes, intermediate values of T show division of labor if $\gamma R \neq 0$. This is described under the section "The role of dominance in task specialization". We added "with intermediate values suggesting a division of labor" to the Figure 2 legend.

(29) L173-175: "as depicted in Figure 2, intermediate values of task specialization indicate in all cases age/dominance-mediated task specialization ($\gamma t \neq 0$; Table 1) and never a lack of specialization ($\gamma t = 0$; Table 1)". This sentence is unclear and imprecise. Does this sentence want to say that in Figure 2, all plots with intermediate values of T involve gamma t different from zero? If so, just say that.

Rephrased to: "In Figure 2, all plots depicting intermediate values of T exhibit non-zero γR values and, hence, division of labor".

(30) L179-180: "forms of help that impact survival never evolve under any environmental condition when only kin selection occurs". This is misleading because under the KS scenario, help cannot positively impact survival in this model, so they never evolve.

Help cannot affect survival but could potentially affect group persistence. If helpers increase breeder productivity and offspring remain philopatric and queue for the breeding position, then they will receive help from related individuals.

(31) L 210: "initially". What do you mean by that?

Help only evolves in our model in family groups, which may then open the door for the evolution of help in mixed-kin groups. Therefore, we use "initially" to refer to the ancestral group structure that likely led to cooperation under benign environmental conditions. We rephrased this section to "in more benign (and often highly productive) environments that lead to habitat saturation, help likely evolved initially in family groups, and defensive tasks are favored because competition for the breeding position is lower under kin selection."

(32) L 212: "kin selection is achieved". What does that mean?

Rephrased to "kin selection acts not only by selecting subordinates in their natal group to increase the productivity of a related breeder [...]"

(33) L 216: "division of labor seems to be more likely to evolve in increasingly harsh environments". Say in parentheses where this is shown.

Added.

(34) L 218: "help evolves in benign environments". I don't see where this is shown. Figure 2 doesn't show that H is higher with lower m (e.g., in KS+GA column).

Help does not evolve in benign environments under only direct fitness benefits derived from group augmentation (shown in Figure 2).

(35) L 225: "y-axis" should be "vertical axis", as y has another meaning in the model.

Done.

(36) L 226: *"likelihood". Here and throughout, "likelihood" should be changed to probability. Likelihood means something else.*

Thank you for the advice, we have corrected this through the manuscript.

(37) L 236: *"the slope of the reaction norm for the dominance value in task specialization".*

Unclear. Clearer to say: the rate at which individuals to shift from defense to work as they age.

The important part is not so much the rate but the direction, that is, from work task to defense (or vice versa) as their rank increases. Changed to “the direction and rate of change in task specialization with dominance”.

(38) L 257: *"(task = 0; cost to dominance value)," This seems out of place.*

This aims to clarify that work tasks have a cost to dominance, while defense tasks have a cost to survival. This is particularly relevant in this model since different helping tasks are defined by their fitness costs.

(39) L 258: *"increase"-> "increase with age".*

Added “with dominance”.

(40) L 262: *"division of labor equilibria" What is that?*

Changed to “at equilibrium when division of labor evolves”

(41) L 268: *"Our findings suggest that direct benefits of group living play a driving role in the evolution of division of labor via task specialization in species with totipotent workers". This is a very general statement, but the results are much more circumscribed. First, the model is quite specific by assuming that, in the absence of group augmentation ($x_n=0$), indirect fitness benefits can only be given to breeders (Equation 5) but not to other subordinates (Equations 2, 3.1). This is unrealistic, particularly for vertebrates, and reduces the possibility that indirect fitness benefits play a role.*

As previously discussed, the scope of this paper was to study division of labor in cooperatively breeding species with fertile workers in which help is exclusively directed towards breeders to enhance offspring production through alloparental care. Other forms of “general” help do not result in task partitioning to enhance productivity.

Second, the difference in costs of work and defense are what drive the evolution of "division of labor" (understood as intermediate T in case this is what the authors mean) in the KS scenario, but the functional forms of those two costs are quite specific and not of the same form, so these functions may bias the results found. Specifically, R is an unbounded linear function of work and the effect of this function becomes weaker as the individual ages due to the weakening force of selection with age (Equation 2) whereas Sh is a particular bounded nonlinear function of defense (Equation 3.1). These differences may tend to make the effect of Sh stronger due to the particular functions chosen.

The difference in costs is inherent to the nature of the different tasks (work versus defense): while survival is naturally bounded, with death as the lower bound, dominance costs are

potentially unbounded, as they are influenced by dynamic social contexts and potential competitors. Therefore, we believe that the model's cost structure is not too different from that in nature.

Third, no parameter sweep is given to see to what extent these results hold across the many parameters involved. So, in summary, the discussion should at least reflect that the results are of a restricted nature rather than giving the impression that they are of the suggested level of generality.

During the exploratory phase of the model development, various parameters and values were assessed. However, the manuscript only details the ranges of values and parameters where changes in the behaviors of interest were observed, enhancing clarity and conciseness. For instance, variation in y_h (the cost of help on dominance when performing “work tasks”) led to behavioral changes similar to those caused by changes in x_h (the cost of help in survival when performing “defensive tasks”), as both are proportional to each other. Specifically, since an increase in defense costs raises the proportion of work relative to defense tasks, while an increase in the costs of work task has the opposite effect, only results for the variation of x_h were included in the manuscript to avoid redundancy. Added to Table 1: “To maintain conciseness, further exploration of the parameter landscape was not included in the manuscript”.

(42) L 270: "in eusocial insects often characterized by high relatedness and reproductive inhibition, sterile workers acquire fitness benefits only indirectly". This is misleading. Sterile workers of any taxa, be it insects or vertebrates, can only acquire fitness benefits indirectly as they are sterile, but eusocial insects involve not only sterile workers.

Rephased to “In contrast, in eusocial species characterized by high relatedness and permanent worker sterility, such as most eusocial insects, workers acquire fitness benefits only indirectly”. In any case, permanent sterility only occurs in eusocial invertebrates; in vertebrates with reproductive inhibition sterility is only temporal and context dependent. Therefore, in vertebrates, sterile workers may potentially obtain direct fitness benefits if the social context changes, as is the case in naked mole-rats.

(43) L 273: "Group members in eusocial species are therefore predicted to maximize colony fitness due to the associated lower within-group conflict". Again, this is incorrect. Primitively eusocial insects have high conflict.

We added “Group members in *such* eusocial species” to clarify that we are not referring here to primitively eusocial species but those with permanent sterile workers.

(44) L 277: "when the benefits of cooperation are evenly distributed among group members". In this model, the benefits of cooperation are not evenly distributed among group members: breeders reproduce, but subordinates don't.

Subordinates may reproduce if they become breeders later in life. However, subordinates also benefit from cooperation as subordinates directly (greater survival in larger groups), and indirectly if they are related to the breeder. Here we refer to the first one, and we expand on that in the following sentence.

(45) L 280: "survival fitness benefits derived from living in larger groups seem to be key for the evolution of cooperative behavior in vertebrates [22, 63], and may also translate into low within-group conflict. This suggests that selection for division of labor in vertebrates is stronger in smaller groups". I don't see how the previous sentence

suggests this. The paper does not present results to support this statement (i.e., no selection gradients in smaller vs larger groups are shown).

The benefits of living in a larger group entail diminishing returns, so those living in smaller groups benefit greater by an increase in productivity and group size than those in a larger group.

(46) L 284: "Our model demonstrates that vertebrates evolve a more stable division of labor". Where is that shown? How is "more stable" measured?

Rephrased to “vertebrates are more likely to evolve division of labor”. This is shown in Figure 2, that exemplifies that division of labor evolves in a wider range of environmental condition and to a higher degree (intermediate values of T).

(47) L 287: "direct fitness benefits in the form of group augmentation select more strongly for defensive tasks". Where is that shown? Establishing this would entail comparing selection gradients with direct fitness benefits of group augmentation and without them.

In Figure 2, when we compare the GA column to KS+GA column, we see that at equilibrium, more helpers choose defense tasks, specially when they are free to choose their preferred task (circles).

(48) L 288: "kin selection alone seems to select only for work tasks." Again, this may be an artifact of the model assuming that helpers cannot increase non-breeders' fitness components except via group augmentation, and that defense tasks are inherently more costly than work tasks.

As stated previously, we are studying task specialization in cooperative breeders where help is in the form of alloparental care (from allofeeding and egg care to defense from predators). We also assume that the costs are different, but whether one or the other is more costly depends on the relative context (e.g., a task can be more costly if it affects competitiveness in a very competitive environment). It is important to note that we name these tasks “work” and “defense” for practical reasons, but the focus of the paper is on tasks with different fitness costs that for their characteristics may not fit so well in under this terminology. While we acknowledge that most tasks have both kinds of fitness costs to a degree, here we focus on the main fitness costs of each kind of task (L430-436).

(49) L 290: "are comparatively large". This sounds as if the tasks are large, which is presumably not what is meant.

Rephrased to “costs to dominance value and to the probability of attaining a breeding position are comparatively larger than survival costs.”

(50) L 298: "helpers are predicted to increase defensive tasks with age or rank, whereas in harsh environments, work tasks are predicted to increase with age or rank." Add parentheses referring to where this is shown.

This is shown in Figure 3, but since this is described in the discussion, we did not add a reference to the figure. If the editor would like us to refer to figures here, we can (see also comments below relating to the same issue).

(51) L 308: "the role of age and environmental harshness on the evolution of division of labor". What is the prediction? Simply, the role of age is an assumption, not a prediction.

Rephrased to “the role of environmental harshness on the evolution of division of labor via age-dependent task specialization”.

(52) L 315: *"individuals shifting from work tasks such as foraging for food, digging, and maintaining the burrow system, to defensive tasks such as guarding and patrolling as individuals grow older and larger". Say in parentheses where this is predicted.*

This prediction comes from Figure 3, we do not reference it here since we are in the Discussion section.

(53) L 320: *"Under these conditions, our model predicts the highest levels of task partitioning and division of labor." Where is this predicted? Add parentheses referring to where this is shown. As it is, it is not possible to check the validity of the statement.*

This prediction comes from Figure 2 column KS+GA, we do not reference it here since we are in the Discussion section. The results with references to the figures are found under the Results section. In the discussion, we reiterate the results already described and add some examples from real data that seem to confirm our predictions.

(54) L 322: *"In line with our model predictions, larger and older helpers of this species invest relatively more in territory maintenance, whereas younger/smaller helpers defend the breeding shelter of the dominant pair to a greater extent against experimentally exposed egg predators". These predictions are neat, but are now very difficult to understand from the figures. Maybe at the bottom of 3A, you could add a diagram work->defense for negative gamma_t and defense>work for positive gamma_t (or whatever order it is).*

Done.

(55) L 325: *"Territory maintenance has been shown to greatly affect routine metabolic rates and, hence, growth rates [80], which directly translates into a decrease in the likelihood of becoming dominant and attaining breeding status, as predicted by our model." This seems to be an assumption, not a prediction.*

That is true. We removed: “as predicted by our model”.

(56) L 352: *"controlled". This means something else.*

Changed to “addressed”.

(57) L 356: *"summary, our study represents the first theoretical model aimed at elucidating the potential mechanisms underlying division of labor between temporal non-reproductives via task specialization in taxa beyond eusocial organisms". Again, claiming to be the first is risky and unnecessary.*

Rephrased to “our study helps to elucidate”.

(58) L 358: *"Harsh environments, where individuals can obtain direct fitness benefits from group living, favor division of labor, thereby enhancing group productivity and, consequently, group size." I'm not sure about this conclusion as harsh environments (large m in Figure 2) also involve the evolution of no division of labor (from the triangles and circles that are zero in the right bottom panel) and perhaps more so than with less harsh environments (intermediate m). Incidentally, in the bottom right panel of Figure 2,*

do the two separate clusters of triangles and circles mean that there is some sort of evolutionary branching?

Yes, there are two different equilibria for the same set of conditions. Although it is true that for $m=0.3$ less division of labor evolves when kin selection and group augmentation act together, it is not the case when only group augmentation takes place. In addition, we qualify $m=0.2$ as harsh as opposed to benign in which we observe the rise of habitat saturation ($m=0.1$). $m=0.3$ is then an extreme harsh environment, in which in several instances different parameter landscape causes population collapse (see figures in the Supplemental Material).

(59) L 360: "Variation in the relative fitness costs of different helping tasks with age favors temporal polyethism". I don't see that this has been shown. Temporal polyethism evolves here whenever γ_t evolves non-zero values. Figure 3A shows that non-zero γ_t evolves with harsher environments, but I don't see what the "variation in relative fitness costs of different helping tasks" refers to.

The evolved reaction norms of the model are towards different fitness costs depending on the task performed, since this is how we define the different types of tasks in the model.

(60) L 382: "undefined". Say variable. Undefined is something else.

Undefined is more accurate, since we did not define how many subordinates there were per group, while “variable” could have been defined within a range, which was not the case in this model.

(61) L 390: "each genetic locus". Say earlier that each genetic trait is controlled by a single locus.

Added.

(62) L 395: "complete" and "consistent" -> "certain".

We changed one to “certain” and another to “absolute” to avoid using the same adjective twice in a sentence.

(63) L 396: What determines whether dispersers become subordinates or floaters? A trait? Or a fixed probability?

We added “which is also controlled by the same genetic dispersal predisposition as for subordinates”.

(64) L 412-413: "cycle". This should be a breeding step.

Changed to “season” instead.

(65) L 418: Say negatively impacts (it could also be positively impacts, which I guess is not what you mean).

Done.

(66) L 425: "a sample of floaters". Chosen how?

Added “randomly drawn”.

(67) L 426-428. But the equation in Table 1 indicates that all floaters compete for breeding spots, not a sample of floaters. This is not clear.

The number of floaters sampled to try to breed at a given group is $N_{f,b} = f \cdot N_f / N_b$ (Table 1).

Therefore, $N_{f,b}$ is the sample size of floaters for a given open breeding position, and f is how many groups on average a floater attempts to access in each time step.

(68) L 432. In the figure, the breeding cycle is called a step, but here it is called a cycle. There should be a single term used throughout. Breeding is not really a cycle here (it doesn't involve multiple steps that are repeated cyclically), so it seems more appropriate to call this breeding steps or breeding seasons.

Taken into account previous comments, we changed the terms “generation” and “life cycle” to “breeding cycle”. We added “or seasons”.

(69) L 439: “generations”. What are generations here, as generations are overlapping? You probably mean time steps or something else.

Changed to “breeding cycles”.

(70) L 439: “equilibrium was reached”. Presumably, equilibrium is reached only asymptotically, so some cutoff is implemented in practice. So maybe say explicitly what cutoff was implemented.

As mentioned, we run the model for 200'000 time steps, and if equilibrium was not reached for the phenotypic values, then we run the model for longer, with 400'000 time steps being the maximum at which all simulation reached equilibrium. In some cases, genetic values did not reach equilibrium at ranges at which there was no impact on phenotypic values, so these were disregarded to assess whether equilibrium was reached.

(71) L 452: “Even though individuals are likely to change the total amount of help given throughout their lives”. Do you mean in real organisms or in the model? Say which. If it is in the model, it is not clear how.

We added “in nature” to clarify that this was not the case in the model.

(72) L 455: “For more details on how individuals may adapt their level of help with age and social and environmental conditions, see [63].” Do you mean real individuals or in the model? Again, if it is in the model, it is unclear how this is possible and should be explained in this paper at least briefly rather than citing another one.

We rephrased it to “How individuals in the model may adapt their level of help with age and social and environmental conditions has been described elsewhere.” We do not go into detail here because it is not within the scope of the paper, and those results have been described elsewhere.

(73) L 475: “helpers”. Make terminology consistent throughout.

All helpers are subordinates, but not all subordinates are helpers, as they may evolve no help. Since here we are describing those subordinates that do help, we use that terminology. We added “subordinate helpers” to clarify this further.

(74) L 476: "proportional". The dependence in Equation 1 is not "proportional to". Say something like "a survival probability (not rate) that decreases with the amount of help provided".

Done.

(75) L 482: "environmental"-> baseline, as defined first.

Done.

(76) L 486: "benefits". Can you briefly say in parentheses what those benefits are in real organisms? As in line 475, where you reminded the reader of survival costs due to predator defense.

Added "such as those offered by safety in numbers or increased resource defense potential".

(77) L 494. "we first outline a basic model in which individuals". It is not clear what this sentence says, and the remainder of this section does not clarify it.

We made two models for comparison, one where individuals can choose freely which task they prefer to perform, and another in which there is an increase in productivity when both kinds of tasks are performed to a similar extent at group level. In the latter model, individuals may choose an unpreferred task at certain times during their lived to increase the effect of the help provided in the breeder's (and group's) productivity.

We rephrased this section to "we first outline a basic model where individuals evolve their preferred helping task. Then we compare this to another model in which the breeder's reproductive outcome is maximized when the group's helping effort in each kind of tasks is performed to a roughly equal degree."

(78) L 496: "by performing both tasks". Sounds as if the breeder performs both tasks, not helpers.

We changed to "when the group's helping effort in each kind of tasks".

(79) L 497: "the maximum amount of cumulative help of each type (σH_{max}) that can affect fecundity is given by Eq. 4." This statement is imprecise. Presumably, what is meant is that this level of help maximises breeder productivity, as stated earlier in the paper. However, there is no proof that this level of help maximises breeder productivity, so this expression seems unjustified and it is unclear how it is used.

This is a description of the model set up. As described later in the same section, the cumulative help of each time that will influence the breeder's fecundity if maximum H_{max} . Therefore, it does represent the maximum amount of cumulative help of each type that can affect the breeder's fecundity.

(80) L 500: "reproduced" -> "reproduce".

Done.

(81) L 503. Say here what K is so that the reader knows what equation 5 is showing.

Added " K " to the "The quantity of offspring produced (K)".

| (82) L 503: "diminishing returns" -> "diminishing returns as help increases".

Done.

| (83) L 507: Why these inequalities?

These inequalities explain the use of H_{max} (response to comment 79). We rephased it to "the cumulative defense effort is larger than $H_{max} (\sum_{i=1}^n H_{defense_i} > H_{max})$ or the cumulative work effort is larger than $H_{max} (\sum_{i=1}^n H_{work_i} > H_{max})$ ".

| (84) L 526: "removing the influence of relatedness from the model". It would be helpful to plot relatedness in this and the other scenario to check that it is indeed low here and high in the other.

The actual values of relatedness are provided in the Supplemental Material Table S1. We added this reference to Figure 2.

| (85) L 528: "It is possible that direct and indirect fitness benefits could have an additive effect on the evolution of alloparental care". This is technically incorrect. It is also unclear what the point of this sentence is.

We have removed this sentence.

| (86) Table 1: Say what are the allowed values for these genotypic traits (can they take negative values, be greater than one, are they continuous or discrete?): e.g., $\alpha \in [0,1]$ or $\alpha \in (-\infty, \infty)$. For phenotypic traits, it would be helpful if the third column lists the equation where the trait is defined. As the variables in the first column are scalars, they should not be bold face. Survival "rate" should be survival "probability" throughout.

All genetic traits can take any real number (-infinity, infinity), but the phenotypic values are either constrained by the equation like for logistic formulas, or manually constrained like for dispersal propensity or help (only positive numbers allowed). We added "Each genetic trait is controlled by a single locus, and may take any real number" (L403), and added the boundaries for help and dominance value in Table 1. We decided against including the equations in the table due to space constraints. We removed the bold face as suggested. We changed all instances of "survival rate" to "survival probability".

| (87) Figures S1, S2: I don't recall seeing references to these figures in the main text, but there should be, as well as for Tables S1-S3.

Table S1 is now referenced in Figure 2. The other figures are now referenced in the main text when we reference the different sections in the Supplemental Materials (L190 and L198). Other Tables are referenced in their respective Figures in the SI.

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